CVT System Model Derivation

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Contents

1 Introduction

1.1 Background and Context

The continuously variable transmission (CVT) is a mechanical power transmission device capable of producing a continuously variable speed ratio between an input and output shaft, in contrast to the fixed discrete ratios of a conventional stepped gearbox. In the class of CVTs relevant to this work — dry rubber V-belt CVTs with mechanically actuated conical sheave pairs — the transmission ratio is not commanded by an external actuator or electronic controller. Instead, it is an emergent dynamic response: the equilibrium of axial forces acting on the primary and secondary sheave pairs, arising from centrifugal flyweight mechanisms, torque-reactive helix and ramp assemblies, and return springs, determines the radial position at which the belt seats within each sheave groove, and therefore the effective pitch radii and transmission ratio at every instant.

This self-regulating character is simultaneously the CVT's greatest practical asset and the primary source of difficulty in modeling it rigorously. Because no actuator explicitly commands the ratio, the ratio trajectory at any instant is the output of a coupled mechanical system whose state depends on engine speed, load torque, belt clamping geometry, and the prior history of the shift position — all evolving together in time.

This class of CVT is found throughout small-displacement powersport and utility vehicles. It is the mandated transmission in the SAE Baja competition series, where a single-cylinder Briggs and Stratton 10 hp engine is paired with a CVT that must be designed, selected, and tuned entirely by student engineering teams from first principles. Beyond Baja, mechanically actuated rubber V-belt CVTs are standard equipment in snowmobiles, all-terrain vehicles (ATVs), utility task vehicles (UTVs), and small agricultural and industrial machinery. Despite their prevalence and the practical engineering importance of understanding their behavior, the quality of publicly available engineering models for these systems is strikingly uneven. The present work was motivated directly by this gap.

1.2 Motivation and Literature Review

1.2.1 Overview of the Modeling Landscape

A thorough review of CVT modeling literature reveals a consistent pattern: each body of work captures some aspects of the physics carefully while leaving equally important aspects absent, empiricized, or explicitly assumed away.

A comprehensive survey of CVT dynamics and control research through 2009 is provided by Srivastava and Haque in their review article *"A Review on Belt and Chain Continuously Variable Transmissions: Dynamics and Control"* Srivastava and Haque (2009).

Their review systematically covers the major CVT types — rubber V-belt, metal pushing V-belt, and chain CVT — across static, quasi-static, and fully dynamic formulations, and surveys both the physical models and the control strategies built upon them. Crucially, it emphasizes that rubber belt CVTs are mechanically and dynamically distinct from metal pushing belt CVTs in their actuation mechanisms, belt compliance, and contact mechanics, and that the two should not be treated interchangeably in modeling.

Their central conclusion — that there exists *"considerable disparity in the type of CVT models that have been used"* across the literature, and that fully dynamic, experimentally validated models remain scarce for several CVT classes — indicates that the field has developed in fragments rather than converging toward a unified framework.

The present section builds on that survey, focusing specifically on the subset of literature relevant to the mechanically actuated rubber V-belt CVT considered in this work.

1.2.2 Metal Pushing V-Belt CVT Models

The most analytically complete CVT dynamic models in the literature concern the Van Doorne-type metal pushing V-belt CVT used in automotive applications.

In this class of transmission, clamping force is imposed hydraulically through electronically controlled oil pressure rather than through centrifugal flyweights and spring mechanisms. The belt itself consists of a series of compressive steel elements rather than a flexible rubber belt operating primarily in tension.

Carbone, Mangialardi, and Mantriota Carbone et al. (2005) derive a dynamic model for the shifting behavior of the metal V-belt CVT and identify two distinct operating regimes governing ratio change. In the first regime, micro-slip redistribution within the contact arc governs the shift process, producing relatively slow ratio evolution. In the second regime, macro-slip occurs and rapid ratio change becomes possible.

Srivastava and Haque extend this class of modeling by incorporating dynamic effects associated with belt element inertia and transient force redistribution within the belt system Srivastava and Haque (2006). These studies represent some of the most detailed dynamic analyses of CVT behavior available in the literature.

Experimental characterization of the slip–torque relationship in metal belt CVTs has also been reported by Bonsen et al. Bonsen et al. (2004). Their results show that transmitted torque varies nonlinearly with slip ratio, increasing through a micro-slip region before reaching a maximum near the transition to macro-slip.

While these models achieve a high degree of dynamic fidelity, they describe a fundamentally different system class from the mechanically actuated rubber belt CVT considered here. The actuation mechanisms, belt mechanics, and contact physics differ substantially, limiting the direct applicability of these formulations.

1.2.3 Rubber Belt Mechanical CVT Models

For the mechanically actuated rubber V-belt CVT, the literature is considerably thinner.

The most detailed dynamic model in the open literature appears to be that of Julio and Plante Julio and Plante (2011). Their formulation discretizes the belt into nodal elements and applies Newton's second law to each node individually. Local friction forces are computed from the normal force at each contact point, allowing the torque transmission capacity of the belt to emerge from the cumulative tangential friction forces acting along the wrap arc.

This nodal formulation has several advantages. The belt tension distribution arises naturally from the equilibrium of individual elements rather than being imposed through the classical capstan relationship. The wedging behavior of the belt in the conical groove is captured directly through the element geometry, and centrifugal effects acting on belt elements are included through fictitious forces in the rotating reference frame.

However, two important limitations remain.

First, the inertia of the movable sheaves is neglected. The axial position of the sheaves is determined through instantaneous force balance rather than through an inertial equation of motion, meaning the model cannot capture the finite time required for sheave motion to respond to changing forces.

Second, the authors report difficulty obtaining stable predictions during the transition between clutching and shifting operation. Since mechanically actuated rubber belt CVTs use the belt clamping mechanism itself as the start-up clutch, the clutching regime forms an essential part of the system operating envelope.

Another related effort is presented in SAE technical paper *"Development of CVT Shift Dynamic Simulation Model with Elastic Rubber V-Belt"* Yi et al. (2011). This work develops a dynamic simulation model incorporating belt elasticity and derives a shift motion relationship based on axial force balance between pulleys. Because the full derivation is not publicly accessible, the detailed structure of the dynamic model cannot be evaluated fully from the available abstract and secondary descriptions.

1.2.4 Quasi-Static Baja-Specific Models

Within the SAE Baja community, most CVT modeling work follows a quasi-static force balance approach.

In these models the operating point of the transmission is obtained by solving the condition that the net axial force acting on the sheave assembly is zero. The shift position is therefore treated as an instantaneous function of operating conditions rather than as a dynamic state evolving over time.

A widely cited example is the thesis by Skinner Skinner (2020), which develops a quasi-static model of the flyweight-actuated primary and ramp-actuated secondary and investigates how parameters such as flyweight mass, spring preload, and ramp angle affect engagement speed and operating RPM.

While such models are useful for design tuning and steady-state analysis, they contain no explicit equation of motion for the shift coordinate, no sheave inertia, and no explicit treatment of belt slip. Consequently they cannot simulate transient behavior or predict the dynamic evolution of the transmission ratio.

1.2.5 The Persistent Gap

The literature survey above reveals a consistent pattern. For mechanically actuated dry rubber V-belt CVTs, no publicly available model simultaneously provides

1. an explicit inertial equation of motion for the axial shift coordinate,
2. first-principles modeling of flyweight actuation,
3. a torque-reactive secondary helix force derived from geometry,
4. centrifugal belt mass correction to effective clamping force,
5. an analytical traction-limit slip criterion, and
6. coupling to vehicle longitudinal dynamics.

Metal pushing-belt CVT models such as those of Carbone et al. (2005), Srivastava and Haque (2006), and Bonsen et al. (2004) achieve high levels of dynamic fidelity but describe a different transmission architecture under hydraulic actuation.

Rubber belt models such as Julio and Plante (2011) capture detailed belt mechanics but omit sheave inertia and encounter difficulties during clutching transitions.

Quasi-static Baja models such as Skinner (2020) provide useful design intuition but cannot capture transient behavior.

The absence of a unified dynamic treatment incorporating both mechanical actuation physics and traction-limited torque transmission motivates the development presented in this work.

Reference	Belt	Actuation	Shift ODE	Sheave Inertia	Clutching Mode	Slip Model	Vehicle Dyn.
Carbone et al. (2005)	Metal	Hydraulic	Yes	Yes	No	Yes	No
Srivastava and Haque (2006)	Metal	Hydraulic	Yes	Yes	No	Yes	No
Bonsen et al. (2004)	Metal	Hydraulic	Partial	Yes	No	Yes	No
Julio and Plante (2011)	Rubber	Mechanical	Partial	No	Unstable	Implicit	No
Yi et al. (2011)	Rubber	Mechanical	Partial	Unknown	Unknown	Not explicit	No
Skinner (2020)	Rubber	Mechanical	No	No	No	No	No
Present Work	Rubber	Mechanical	Yes	Yes	Yes	Analytical	Yes

Table 1: Comparison of representative CVT modeling approaches

1.3 Objectives and Scope

This work has two parallel objectives.

The first is technical: to derive a complete, coupled system of nonlinear ordinary differential equations governing the CVT from engine input shaft to vehicle wheel, suitable for direct numerical integration in time.

The second is pedagogical: to make each step of the derivation explicit and physically interpretable, allowing the reader to develop intuition for the mechanisms governing CVT behavior.

The model includes:

- Primary sheave axial shift dynamics
- Secondary helix and spring force model
- Belt geometry and kinematics
- Belt traction and slip-state model
- Rotational equations of motion for both pulleys
- Vehicle longitudinal dynamics

The model intentionally excludes thermal effects, belt wear, detailed tribological contact mechanics, and multi-body vehicle dynamics beyond one-dimensional longitudinal motion.

1.4 Contribution Statement

The primary technical contributions of this work are:

1. A unified nonlinear state-space model for the mechanically actuated rubber V-belt CVT.
2. First-principles derivation of the primary axial shift equation of motion including sheave inertia and centrifugal flyweight actuation.

3. First-principles derivation of the torque-reactive secondary helix force.
4. Derivation of the centrifugal belt mass correction to effective clamping force.
5. An analytical slip-state switching criterion separating no-slip and traction-limited regimes.
6. Full coupling of the CVT dynamics to vehicle longitudinal motion.

1.5 Document Organization

The remainder of this document is organized as follows.

Section 2 describes the physical CVT system and its components. Section 3 defines notation and symbol conventions. Section 4 defines the dynamic state variables.

Sections 7 through 11 contain the core derivations: pulley geometry, primary shift dynamics, secondary force models, belt traction mechanics, and the slip-state formulation.

Section 12 assembles the complete dynamic system, and Section 13 introduces the vehicle longitudinal dynamics module.

2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the fundamental operating principles of a continuously variable transmission (CVT) and then presents the specific CVT and drivetrain architecture considered in this work.

The goal is to establish clear mechanical intuition before developing the mathematical model in later sections.

2.1 Geometric Principle of a Belt-Sheave CVT

A belt-type CVT consists of two pulleys connected by a flexible belt. Each pulley is formed by two opposing conical faces (sheaves). The axial separation between these sheaves determines the effective radius at which the belt rides.

When the sheaves move closer together, the belt is forced outward to a larger radius. When the sheaves move apart, the belt settles at a smaller radius. Because the belt length remains approximately constant, an increase in effective radius on one pulley necessarily causes a corresponding decrease on the other.

As a result, the transmission ratio changes continuously as the pulley geometries change. Unlike a traditional gearbox, there are no discrete gear steps; the ratio evolves smoothly between its minimum and maximum values.

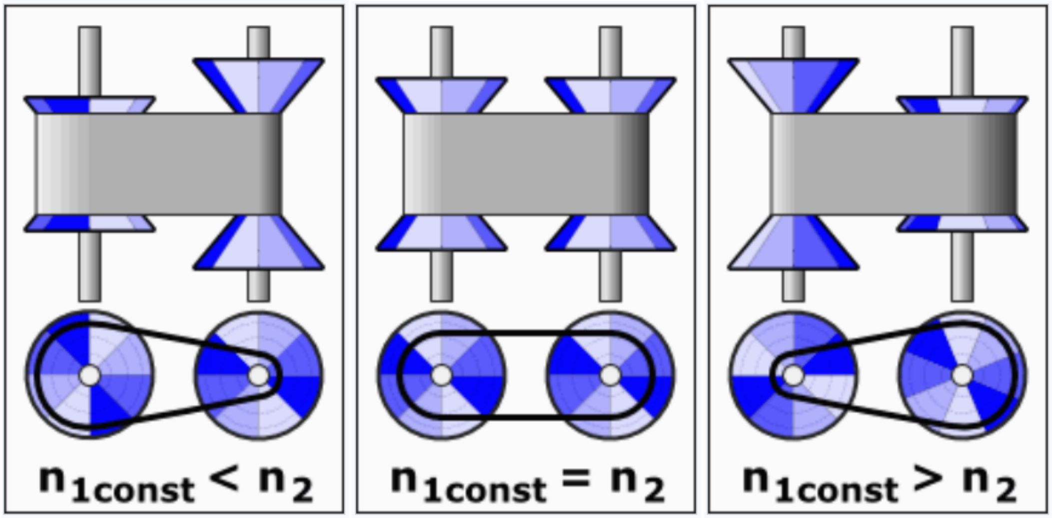

Figure 1: Conceptual illustration of a belt-sheave CVT at three different ratios. Changing the axial separation of the conical sheaves alters the effective belt radius, thereby modifying the transmission ratio continuously.

2.2 Continuous Ratio Versus Discrete Gearing

The defining feature of a CVT is that its transmission ratio can vary smoothly rather than in discrete steps. To understand why this matters, it is useful to first consider how a conventional transmission behaves.

In a stepped gearbox, each gear pair imposes a fixed relationship between engine speed and wheel speed. When vehicle speed increases beyond the useful range of one gear, the transmission must shift to another. This produces an abrupt change in engine speed for the same vehicle speed. As a result, the engine repeatedly accelerates and decelerates as the vehicle progresses through the gears.

A belt-sheave CVT operates differently. Because the effective pulley radii change continuously, the speed relationship between the engine and drivetrain can be adjusted smoothly at any instant. There are no fixed gear pairs that constrain the system to specific ratios. Instead, the ratio can evolve gradually as operating conditions change.

This continuous adjustability allows the engine to operate near a preferred speed while the vehicle accelerates. For example, if an engine produces its peak power within a narrow speed band, a CVT can maintain the engine near that region while vehicle speed increases steadily. Rather than stepping through gears, the transmission adapts its ratio to match the engine's most effective operating condition.

The result is smoother acceleration, improved power utilization, and greater flexibility in matching engine behavior to load demands.

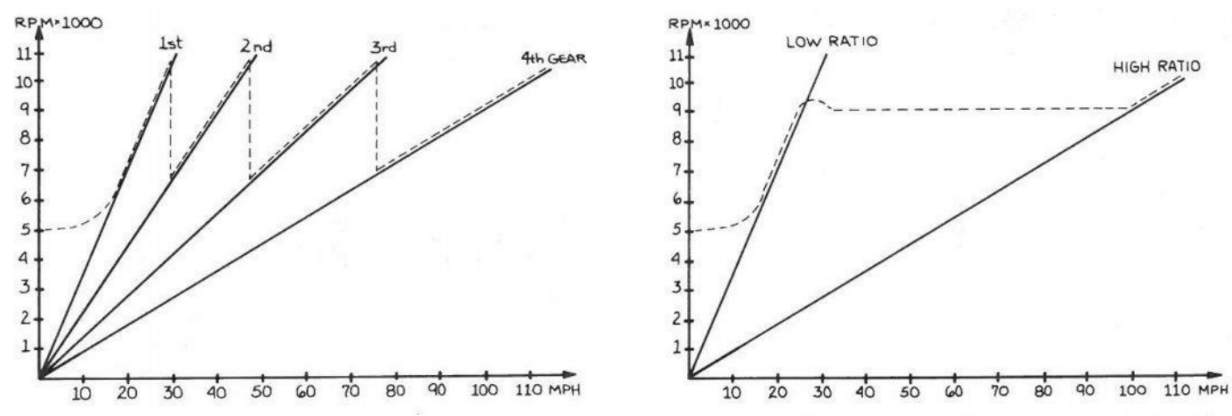

Figure 2: Illustrative comparison of engine speed versus vehicle speed. Left: stepped transmission showing discrete shifts. Right: continuously variable transmission showing smooth ratio evolution.

2.3 Traction-Based Torque Transmission

While the geometric adjustment of pulley radii determines the speed ratio, torque is transmitted through frictional interaction between the belt and the sheave faces.

When a pulley attempts to rotate, it applies a tangential force to the belt. For that force to be transmitted without relative motion, sufficient normal clamping force must press the belt against the conical surfaces. The greater the normal force, the greater the tangential force that can be supported before slip occurs.

This requirement arises directly from the nature of friction. The belt does not contain rigid gear teeth that mechanically lock the pulleys together. Instead, torque transfer depends on traction generated by contact pressure and surface friction. If the transmitted torque exceeds the available traction capacity, the belt will begin to slip relative to the sheave faces.

Slip reduces efficiency, generates heat, and can accelerate component wear. For this reason, CVT designs incorporate mechanisms that actively regulate clamping force in response to operating conditions.

It is therefore useful to distinguish between two interacting aspects of CVT behavior:

- The **geometric ratio**, which determines relative speeds,
- The **traction capacity**, which determines how much torque can be transmitted without slip.

Both must be considered to fully understand performance.

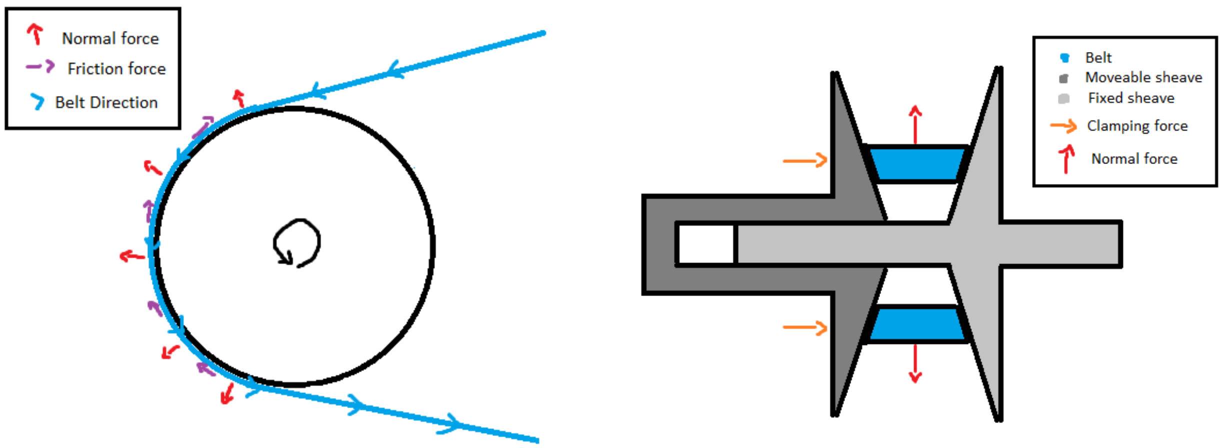

Figure 3: Belt-pulley contact mechanics and force transmission. Left: belt wrapped around a pulley showing normal forces (pressing belt against sheave surface) and tangential friction forces (transmitting torque through traction). Right: cross-sectional view of a CVT pulley assembly showing how axial clamping force between opposing sheaves generates the normal force required for friction and torque transmission.

2.4 Actuation Mechanisms in the Present CVT

While the geometric principles described above apply to all belt-type CVTs, the manner in which axial motion and clamping forces are generated depends on the specific hardware implementation.

The CVT considered in this work employs:

- A **centrifugal flyweight and ramp mechanism** on the primary pulley,
- A **torque-feedback helix mechanism** on the secondary pulley,
- Compression and torsional springs that regulate axial clamping forces.

On the primary side, rotating flyweights move outward under centrifugal effects. Through a ramped interface, this outward motion generates axial force, which adjusts the sheave separation and alters the effective radius.

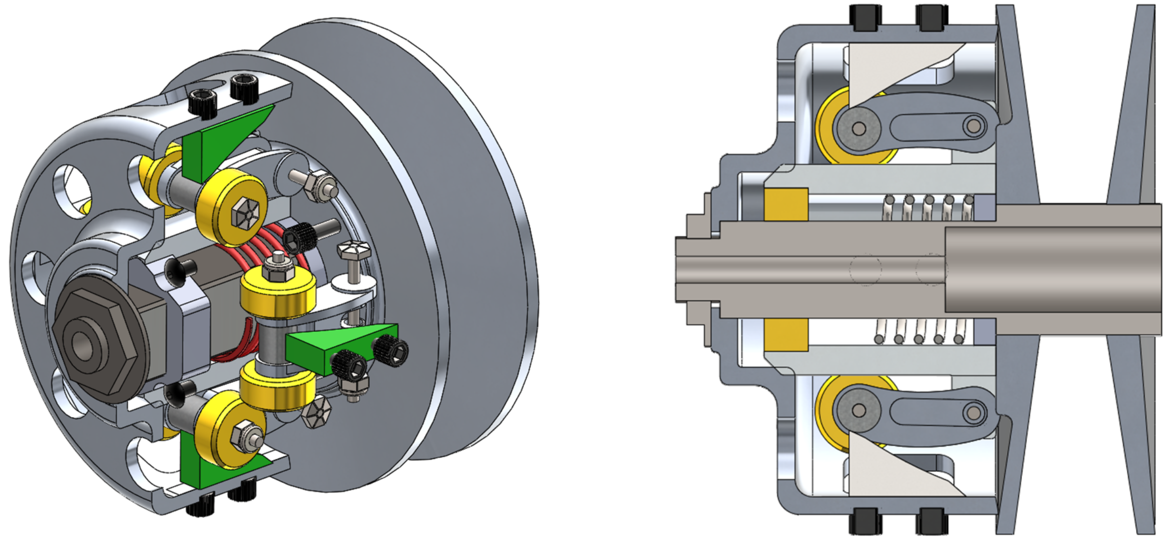

Figure 4: Primary CVT pulley assembly. Left: exterior view. Right: cross-sectional view showing centrifugal flyweights (yellow), ramps (green), and compression spring (red). As engine speed increases, centrifugal force drives the flyweights outward along the ramps, generating axial clamping force.

On the secondary side, transmitted torque produces a rotational reaction within a helical interface. This interaction generates axial force that increases clamping in response to torque demand. Springs supplement this mechanism to regulate baseline clamping and system stability.

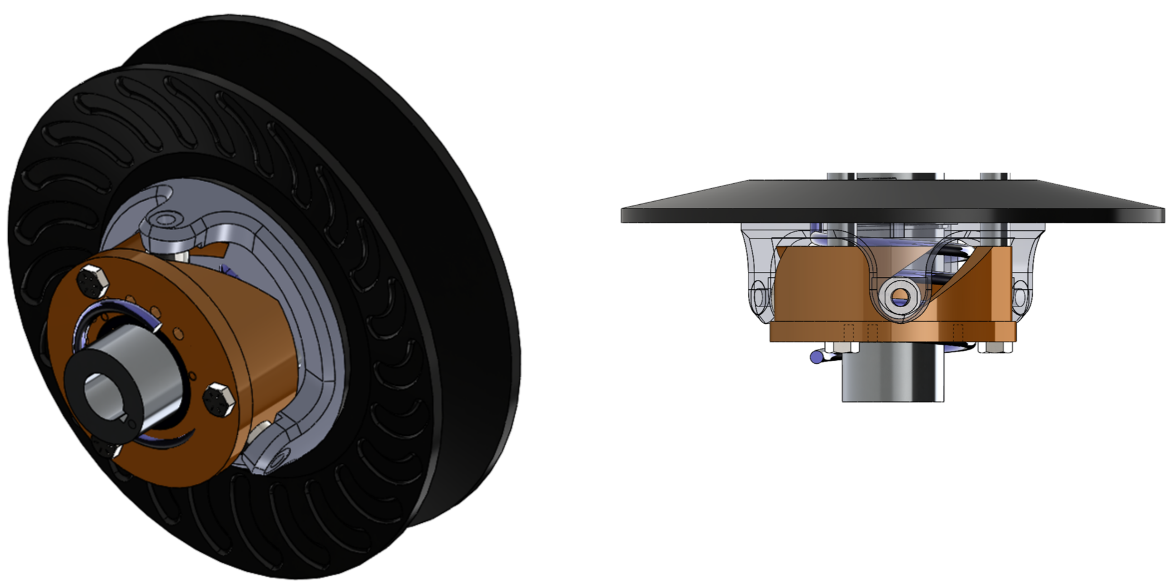

Figure 5: Secondary CVT pulley assembly. Left: exterior view. Right: cross-sectional view showing torque-feedback helix mechanism (bronze) and torsional spring (black). Transmitted torque generates a rotational reaction in the helix, which produces axial clamping force proportional to load.

Although these mechanisms are specific to the hardware used in this system, the underlying principles of variable effective radius and traction-based torque transmission remain common to belt-driven CVTs.

2.5 Full Drivetrain Architecture

The CVT does not operate in isolation. In the system considered here, the drivetrain consists of:

- An internal combustion engine,
- The primary CVT pulley,
- A belt connecting to the secondary CVT pulley,
- A fixed-ratio gearbox,
- Final drive components and driven wheels,
- Tire-ground interaction generating tractive forces.

Engine torque enters the primary pulley, is transmitted through the belt to the secondary pulley, passes through a constant gear reduction, and ultimately produces tractive force at the ground.

The interaction between engine dynamics, CVT ratio evolution, drivetrain inertia, and external load forces governs overall vehicle performance.

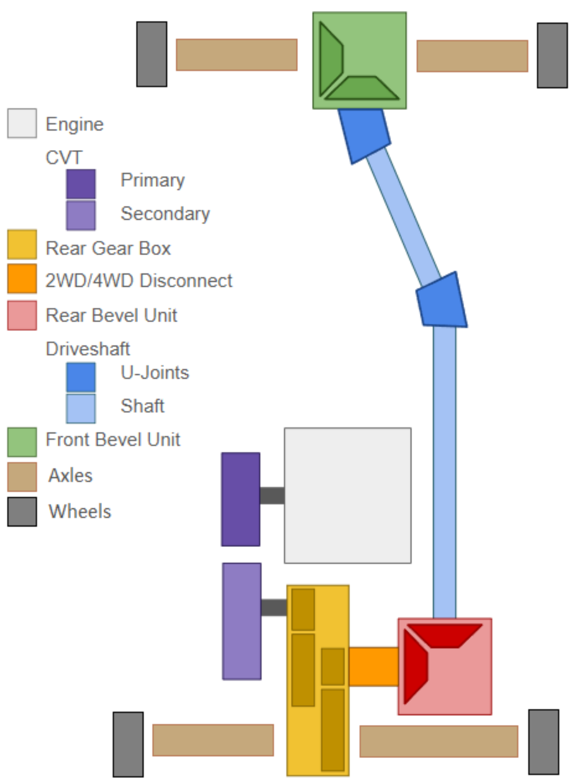

Figure 6: Overview of the complete drivetrain architecture considered in this work.

3 Reference Material

This section records any information for easy reference.

3.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

3.1.1 Vehicle Longitudinal Coordinate and Incline Angle

The vehicle is modeled as moving along a single straight line aligned with the road surface.

The positive longitudinal direction, denoted $+x$, is defined as the forward direction of travel along the surface of the road.

The road may be inclined relative to horizontal. The incline angle is denoted by α and is measured between the road surface and the horizontal reference line. An incline with $\alpha > 0$ corresponds to the vehicle facing uphill in the positive x direction.

Importantly, the $+x$ axis is *tangent to the road surface* and rotates with the road inclination. The $+x$ direction is not fixed in space but rather follows the path of the road, changing orientation as the incline angle α changes.

Because the $+x$ axis remains parallel to the road surface regardless of incline, the coordinate system rotates through angle α relative to a horizontal reference frame.

A vehicle with velocity $v > 0$ is therefore moving forward along the road in the $+x$ direction. If $v < 0$, the vehicle is moving in the opposite direction.

These definitions establish the geometric orientation of vehicle motion and the road surface.

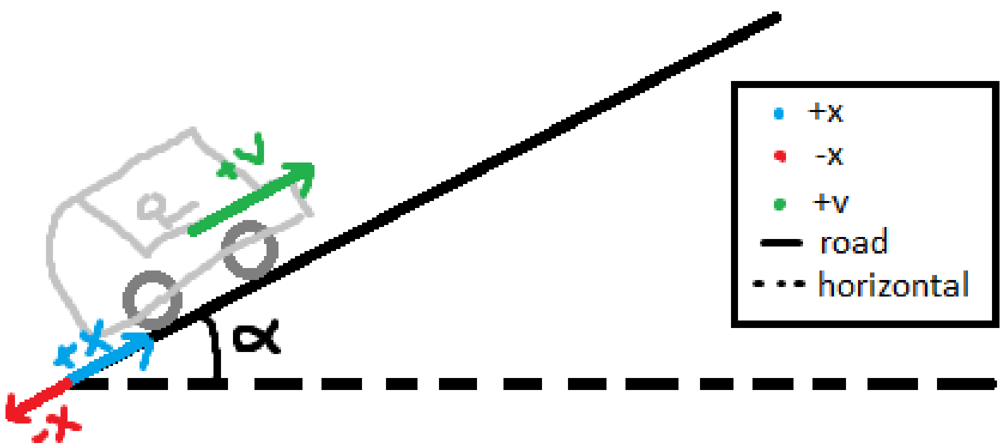

Figure 7: Definition of the longitudinal coordinate and incline angle. The positive x direction follows the road surface. Velocity v is positive when directed uphill along $+x$. The incline angle α is measured between the road and the horizontal.

3.1.2 Rotational Sign Convention

All rotating components in the drivetrain follow a consistent angular sign convention.

The primary pulley, secondary pulley, and engine crankshaft are each assigned a positive direction of rotation. This positive rotational direction is defined such that it produces forward belt motion and, consequently, forward vehicle motion along the $+x$ direction defined previously.

Angular velocities are therefore interpreted as follows:

- $\omega_p > 0$ corresponds to the primary pulley rotating in the direction that drives the belt forward.
- $\omega_s > 0$ corresponds to the secondary pulley rotating in the direction consistent with forward belt travel.
- Engine angular velocity is positive when rotating in the direction that drives the primary pulley forward.

Torque follows the same convention. A torque is considered positive if it acts in the defined positive direction of rotation and tends to increase the corresponding angular velocity. A torque acting opposite to the defined rotational direction is negative.

These definitions establish a consistent orientation for all rotational quantities in the drivetrain.

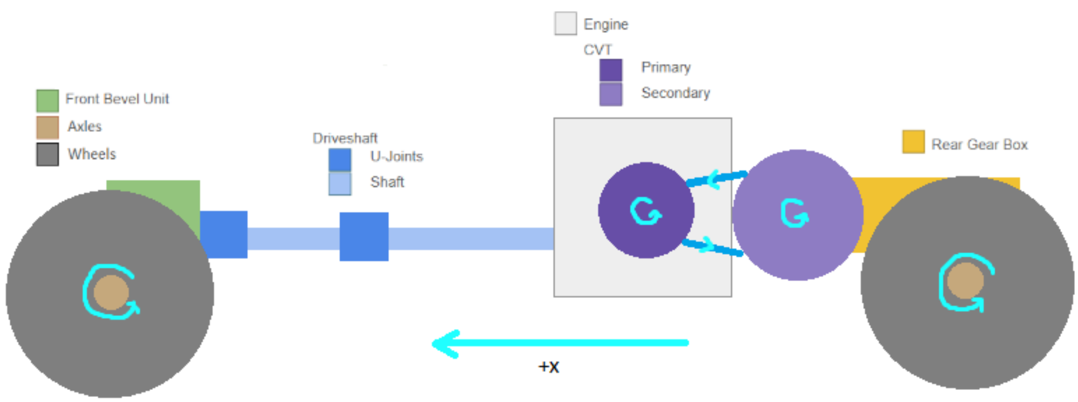

Figure 8: Rotational sign convention for the drivetrain. Positive angular velocities (ω_p, ω_s) correspond to rotation that drives the belt in the $+x$ direction, producing forward vehicle motion. Positive torque acts in the same rotational direction.

3.1.3 CVT Axial Direction

The axial direction of the primary pulley is defined along the axis of its shaft. This direction is parallel to the pulley centerline and perpendicular to the belt's plane of rotation.

The primary pulley is initially in an open configuration, meaning there is a finite gap between the fixed and movable sheaves. The axial coordinate s does not represent the total gap distance. Instead, it measures how much the movable sheave has translated from this initial open position.

A positive axial displacement, $s > 0$, corresponds to the movable sheave moving toward the fixed sheave, thereby reducing the gap. In other words, increasing s represents additional closure relative to the initial open state.

As the sheaves move closer together, the belt is forced outward along the conical faces, increasing the effective belt radius on the primary pulley. Thus, increasing s corresponds to increasing primary effective radius and produces an upshift.

A change in s also affects the secondary pulley. Because the belt length remains constrained, motion of the primary sheaves is coupled to motion of the secondary sheaves. In general terms, an increase in s on the primary corresponds to the secondary sheaves moving apart. The precise relationship between these motions depends on belt geometry and will be established later; here it is sufficient to recognize that the two axial motions are linked.

Axial velocity follows the same sign convention: $\dot{s} > 0$ corresponds to the movable sheave translating in the defined positive axial direction.

These definitions establish the orientation and meaning of axial motion for the CVT.

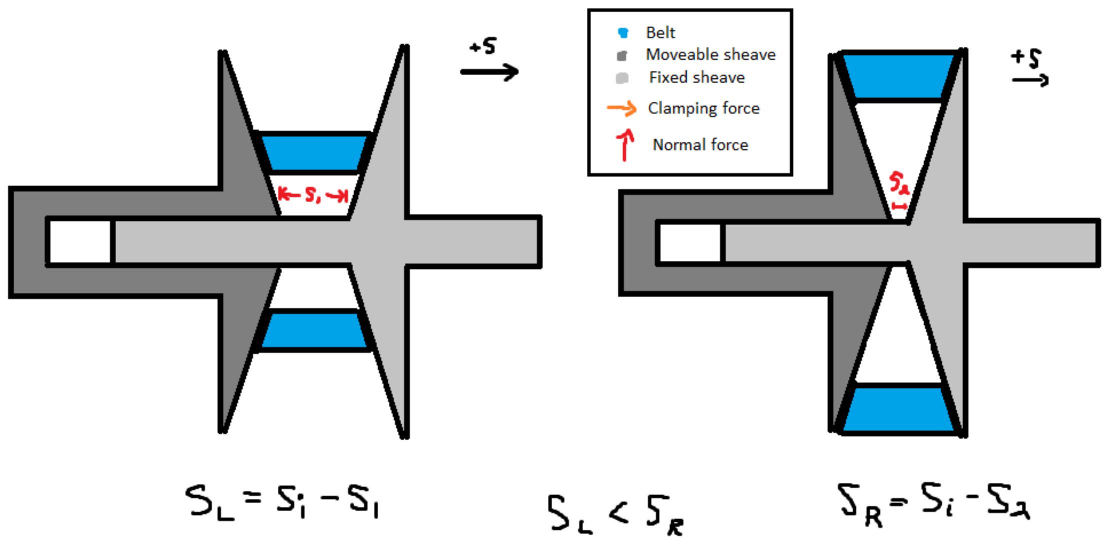

Figure 9: Definition of the primary axial coordinate. The system begins in an open configuration. The axial displacement s measures how much the movable sheave has closed from this initial position. Increasing s increases primary effective radius and is coupled to secondary axial motion.

3.1.4 Radial and Tangential Directions

Within the plane of pulley rotation, directions are defined using the standard polar (radial–tangential) convention commonly used in rotational mechanics.

The radial direction is defined perpendicular to the shaft axis and measured outward from the center of rotation. At any point on the pulley, the radial direction points directly away from the center toward the belt contact surface.

A radial force is defined as positive when it acts outward from the center of the pulley. Thus, $F_r > 0$ corresponds to a force directed away from the center.

The tangential direction lies in the plane of rotation and is perpendicular to the radial direction at each point. The positive tangential direction is defined to be consistent with the previously defined positive angular rotation. In other words, positive tangential direction corresponds to the direction of motion produced by positive angular velocity defined in Section 3.1.2.

Together, the radial and tangential directions form the standard local coordinate basis for forces and motion in the rotating pulley.

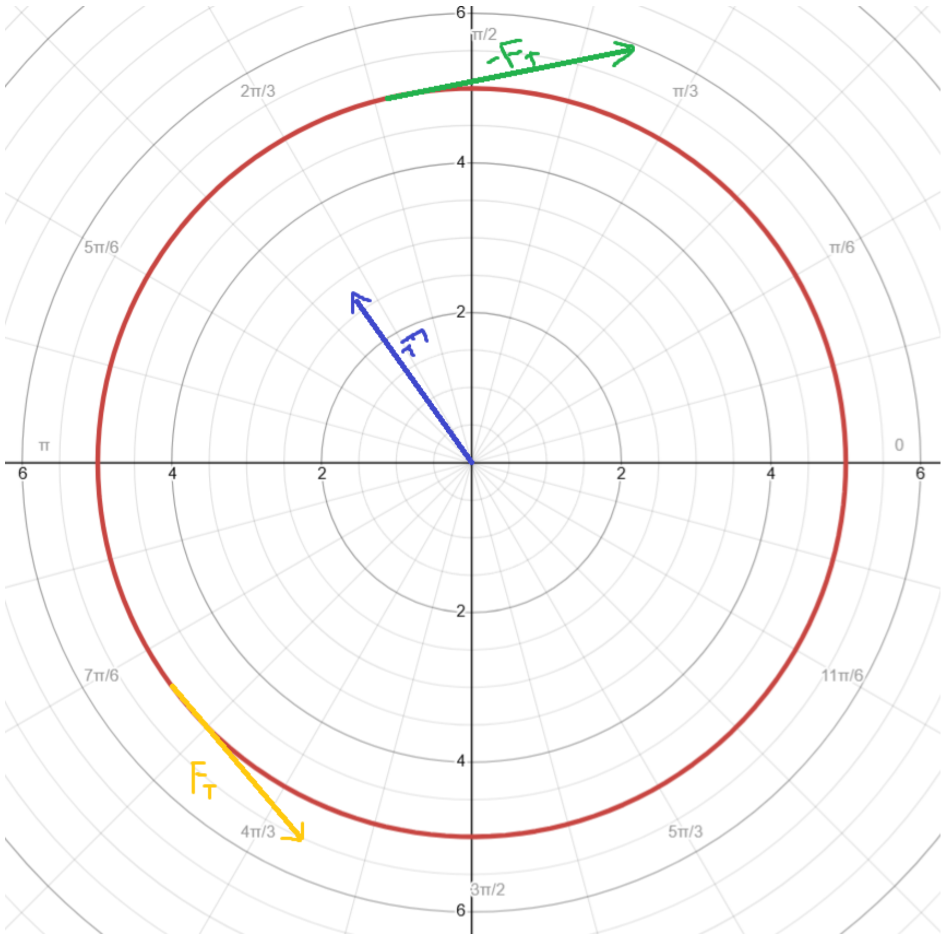

Figure 10: Definition of radial and tangential directions in the plane of rotation. The radial direction points outward from the center. The tangential direction is perpendicular to the radial direction and aligned with positive rotation.

3.2 Symbol Summary

Symbol	Meaning
v	vehicle speed
ω_e	engine angular speed
s	primary axial shift position
$\dot{s}$	primary axial shift velocity
R	CVT ratio (define direction explicitly)

Table 2: Core symbols used throughout the derivation.

3.3 Base Units

All quantities in this document are expressed in SI units unless otherwise stated.

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Plane angle	radian	rad

Table 3: SI base units used throughout the model.

3.4 Common Derived Units

Quantity	Unit	Symbol
Velocity	meter per second	m/s
Acceleration	meter per second squared	$\mathrm{m/s^2}$
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	$\mathrm{rad/s^2}$
Force	newton ($\mathrm{kg\,m/s^2}$)	N
Torque	newton meter ($\mathrm{kg\,m^2/s^2}$)	N m
Moment of inertia	kilogram meter squared	$\mathrm{kg\,m^2}$
Power	watt ($\mathrm{kg\,m^2/s^3}$)	W

Table 4: Common derived SI units used in the CVT model.

3.5 Constants and Parameters

All physical constants, geometric parameters, and calibrated values used in the model are listed below.

Symbol	Name	Value	Unit	Description
m	Vehicle mass	–	kg	Total vehicle mass
r_w	Wheel radius	–	m	Effective rolling radius
G	Final drive ratio	–	–	Secondary-to-wheel speed ratio
I_p	Primary inertia	–	$\mathrm{kg\,m^2}$	Lumped primary-side inertia
I_s	Secondary inertia	–	$\mathrm{kg\,m^2}$	Lumped secondary-side inertia
k_p	Primary spring constant	–	N/m	Primary compression spring
k_s	Secondary spring constant	–	N/m	Secondary compression spring
μ	Friction coefficient	–	–	Belt-pulley friction coefficient
β	Wrap angle	–	rad	Belt wrap angle

Table 5: Model constants and parameters.

4 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations remain valid.

4.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

4.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified friction law (e.g., Coulomb friction with a capstan formulation and an explicit stick–slip transition rule). Microscopic contact mechanics, rubber hysteresis, lubrication films, and detailed Stribeck effects are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, or surface conditioning.

Assumption 6 (Uniform Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

4.3 Geometric and Kinematic Idealizations

Assumption 7 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 8 (Constant Belt Length)

The belt length is treated as constant. Permanent creep and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

Assumption 9 (Uniform Contact Radius Around Wrap)

At any given instant, the belt contacts each pulley at a single, uniform effective radius throughout the entire wrap angle. The contact radius is constant around the circumference of contact, with no spiral or progressive radial variation along the wrap. While the effective radius may evolve with axial shift position, it remains spatially uniform at each time instant.

4.4 Dynamic and Inertial Modeling

Assumption 10 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 11 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 12 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

Assumption 13 (Ideal CVT Power Transmission)

Power transmission through the CVT is modeled as ideal (i.e., no internal dissipation). Although belt slip is explicitly modeled and affects torque capacity and shift dynamics, the associated power loss P_{loss} is neglected.

4.5 Environmental and Operational Scope

Assumption 14 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 15 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

5 Governing Principles

Theory 1: Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s^2)

Source

Classical mechanics (textbook / lecture notes).

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Theory 2: Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia ($\mathrm{kg\,m^2}$)
- α is angular acceleration ($\mathrm{rad/s^2}$)

Source

Classical rigid body dynamics (textbook / lecture notes).

Derivation

Standard rigid-body rotational dynamics relationship.

Theory 3: Torque Definition

Equation

$$\tau = Fr$$

Description

- τ is torque (N m)
- F is tangential force (N)
- r is moment arm radius (m)

Source

Classical mechanics (textbook / lecture notes).

Derivation

Defines torque as the product of tangential force and moment arm. Equivalently, $F = \tau/r$ gives the tangential force from a known torque.

Theory 4: Kinetic Energy (Translational)

Equation

$$T = \frac{1}{2}mv^2$$

Description

- T is kinetic energy (J)
- m is mass (kg)
- v is velocity (m/s)

Source

Classical mechanics (textbook / lecture notes).

Derivation

Translational kinetic energy of a point mass or rigid body moving with velocity v.

Theory 5: Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

Bogna Szyk and Steven Wooding (2024).

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Theory 6: Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

Encyclopaedia Britannica (2024).

Derivation

Linear spring constitutive relationship in the elastic regime.

Theory 7: Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

Dr. Templeman (2024).

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Theory 8: Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

2024 Power Test, LLC (2024).

Derivation

From the rotational power relationship $P = \tau\omega$.

Theory 9: Gearing (Torque Scaling)

Equation

$$\tau_{\text{out}} = \tau_{\text{in}}\, R_{\text{gear}}$$

Description

- τ_{out} is output torque (N m)
- τ_{in} is input torque (N m)
- R_{gear} is gear reduction ratio (unitless)

Source

Brain (2023).

Derivation

Ideal gear relationship assuming no losses; power balance implies inverse scaling of speed.

Theory 10: Coulomb Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

Wikipedia contributors (2024).

Derivation

Coulomb friction model for impending slip.

Theory 11: Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

Hackaday (2021).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{\mathrm{d}T}{T} = \mu \, \mathrm{d}\theta$, then integrate over wrap.

Theory 12: Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density (kg/m^3)
- m is mass (kg)
- V is volume (m^3)

Source

The Editors of Encyclopaedia Britannica (2024).

Derivation

Definition of density.

Theory 13: Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m^3)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m^2)

Source

SoftSchools (2020).

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

Theory 14: Rolling Resistance

Equation

$$F_r = C_r N$$

Description

- F_r is rolling resistance force (N)
- C_r is coefficient of rolling resistance (unitless)
- N is normal force (N)

Source

?.

Derivation

Rolling resistance arises from deformation of contact surfaces (tire and road). For a vehicle on level ground, $N = mg$ where m is vehicle mass and g is gravitational acceleration.

Theory 15: Gravitational Force

Equation

$$F_g = mg$$

Description

- F_g is gravitational force (N)
- m is mass (kg)
- g is gravitational acceleration (m/s^2)

Source

?.

Derivation

Near Earth's surface, $g \approx 9.81$ m/s^2. On an incline with angle θ, the component parallel to the surface is $F_g \sin\theta$.

6 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables.* A state variable is a quantity whose future evolution is determined by its current value and the governing differential equations of the system.

In numerical simulation, state variables are the quantities that are explicitly integrated forward in time. All other model quantities — such as torque transfer, transmission ratio, belt forces, and contact conditions — are computed algebraically from these states.

6.1 State Vector

State Vector **(6.1)**

$$\boldsymbol{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \end{bmatrix}$$

where:

- ω_p is the angular speed of the primary pulley (rad/s),
- ω_s is the angular speed of the secondary pulley (rad/s),
- s is the axial shift position of the primary sheave (m),
- $\dot{s}$ is the axial shift velocity (m/s).

6.2 Independent Dynamic Degrees of Freedom

These four quantities represent the independent dynamic motions of the CVT system.

The primary pulley can rotate independently under engine input and belt interaction. Its angular speed ω_p therefore represents one rotational degree of freedom.

The secondary pulley also rotates and interacts with both the belt and the external drivetrain load. Its angular speed ω_s represents a second independent rotational degree of freedom.

In addition to rotation, the CVT includes axial motion of the primary sheave. This axial displacement s determines the effective pulley geometry and therefore the transmission ratio. The axial motion is mechanically independent of pure rotation and must be modeled explicitly.

Because axial motion follows second-order dynamics, both the axial position s and its rate $\dot{s}$ are required to fully describe its evolution.

Together, these variables capture the complete rotational and axial motion of the system. Removing any one of them would eliminate an independent dynamic degree of freedom from the model.

6.3 Time Integration Perspective

During numerical simulation, the state vector is advanced forward in time.

At each timestep Δt:

- The angular speeds ω_p and ω_s are updated according to their angular accelerations.
- The axial position s is updated using its current velocity $\dot{s}$.
- The axial velocity $\dot{s}$ is updated according to the axial acceleration.

The angular and axial accelerations are determined from the governing equations derived in subsequent sections. Once the state vector has been updated, all other quantities in the model are computed directly from the current values of ω_p, ω_s, s, and $\dot{s}$.

7 Pulley Geometry

This section establishes the geometric relationships that govern how axial shift modifies the effective pulley radii and therefore the transmission ratio.

The analysis presented here is purely kinematic. No force balances or dynamic effects are considered. The goal is to describe how pulley geometry alone determines belt position and ratio evolution.

The objective is to derive explicit expressions for:

- Primary outer radius as a function of shift,
- Secondary outer radius as a function of shift,
- Center-to-center distance,
- CVT ratio and its rate of change.

7.1 Conceptual Overview of Variable Geometry

As introduced in Section 2, each pulley consists of two opposing conical sheaves. One sheave is fixed axially, while the other translates along the shaft.

The axial separation between the sheaves determines where the belt contacts the conical faces. When the movable sheave translates toward the fixed sheave, the gap decreases and the belt is forced outward, increasing the effective pulley radius. When the sheaves separate, the belt settles deeper between them, decreasing the effective radius.

It is important to distinguish between axial displacement and effective radius. Axial motion changes the gap between sheaves; the effective radius changes as a geometric consequence of how the belt sits within that gap. The relationship between these quantities will be derived explicitly in the following subsection.

Because the belt length remains constrained, the effective radii of the primary and secondary pulleys cannot vary independently. An increase in the primary effective radius must be accompanied by a corresponding decrease in the secondary effective radius. This geometric coupling is fundamental to CVT operation and forms the basis of the ratio model derived below.

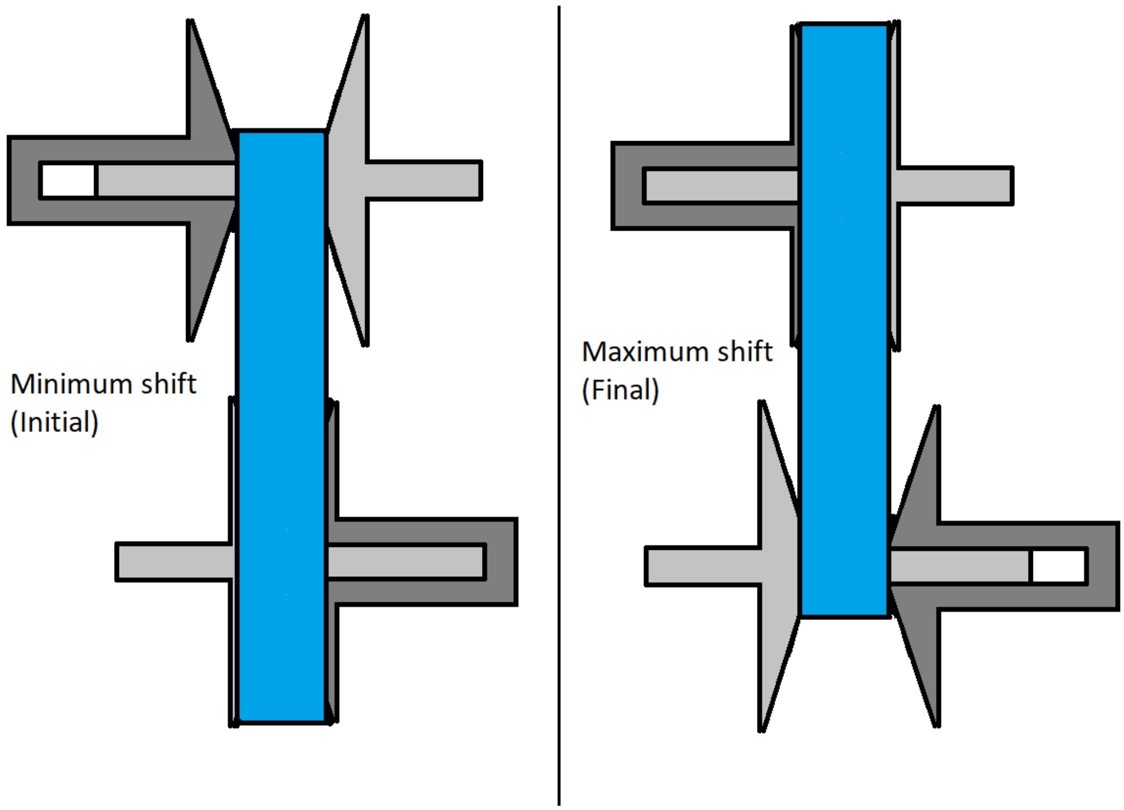

Figure 11: Conceptual illustration of variable pulley geometry. Changes in axial separation alter effective radii. Because belt length is constrained, an increase in primary effective radius corresponds to a decrease in secondary effective radius.

7.2 Definition of Axial Shift Variable

Let s denote the axial displacement of the movable primary sheave, defined according to the sign convention in Section 3.1.3.

At $s = 0$, the primary pulley is in its fully open configuration. The sheaves are at their maximum axial separation, and the belt rests at its minimum effective radius.

The shift coordinate s measures the amount of closure relative to this initial configuration. Increasing s corresponds to the movable sheave translating toward the fixed sheave, reducing the axial gap.

Deadzone and Travel Limits In practice, the initial sheave gap at $s = 0$ is larger than the belt width. As a result, small axial motion does not immediately alter the belt's contact position on the conical faces.

The interval

$$0 \leq s < s_{\mathrm{dz}}$$

defines this *deadzone*. Within this region, the movable sheave translates axially but does not yet sufficiently compress the belt to change its effective contact radius. The belt remains at the same radial position despite the reduction in gap.

Once the sheave gap becomes small enough to fully engage the belt, further axial motion alters the belt position. For

$$s_{\text{dz}} \leq s \leq s_{\max},$$

additional closure forces the belt outward along the conical faces, producing a proportional change in primary effective radius.

The shift coordinate is therefore bounded as

$$0 \leq s \leq s_{\max}.$$

The upper limit $s_{\max}$ corresponds to the physical travel limit of the sheave assembly. At this position, the sheaves are fully closed and mechanically contact one another, forming a hard geometric boundary beyond which further axial motion is not possible.

7.3 Primary Outer Radius as a Function of Shift

When the primary sheaves close beyond the deadzone, the belt is forced outward along the conical faces. The relationship between axial closure and radial expansion follows directly from the cone geometry.

As shown in Figure 12, the sheave face forms a straight line in cross-section. The cone half-angle β is defined as the angle between the sheave face and the *radial direction* (As defined in Section 3.1.4).

For a small axial displacement Δs beyond the deadzone, the total closure is shared symmetrically between the two opposing sheaves, so each sheave contributes $\Delta s/2$.

From the right triangle formed by the sheave face,

$$\tan\beta = \frac{\Delta s/2}{\Delta r},$$

which gives

$$\Delta r = \frac{\Delta s}{2\tan\beta}. \tag{7.1}$$

Thus, the cone angle determines how axial motion is converted into radial growth: smaller β produces greater radial change for the same axial displacement.

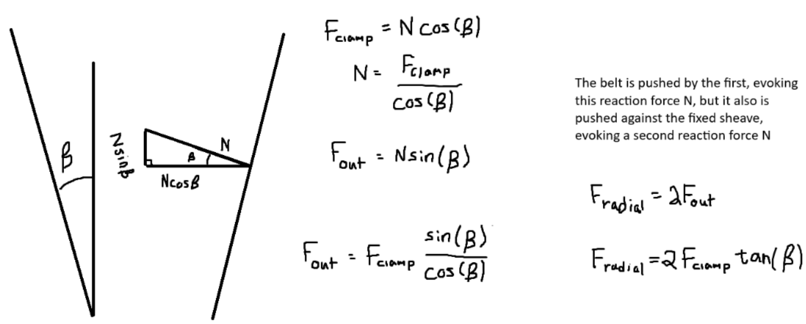

Figure 12: Axial–radial relationship for opposing conical sheaves. The angle β is defined between the sheave face and the radial direction.

Piecewise Radius Model Let $r_{p,\min}$ denote the minimum primary outer radius, corresponding to the fully open configuration ($s = 0$).

Within the deadzone, axial motion does not alter belt position. Once $s \geq s_{\mathrm{dz}}$, the geometric relation above applies, yielding

Primary Outer Radius **(7.2)**

$$r_{p,\mathrm{outer}}(s) = \begin{cases} r_{p,\min}, & 0 \leq s < s_{\mathrm{dz}}, \\ r_{p,\min} + \dfrac{s - s_{\mathrm{dz}}}{2\tan\beta}, & s_{\mathrm{dz}} \leq s \leq s_{\max}. \end{cases}$$

7.4 Belt Length Constraint

In the previous subsection, the primary effective radius was expressed as a function of axial shift. The secondary radius, however, cannot be chosen independently. The belt has a fixed total length, and therefore any change in primary radius must be accompanied by a corresponding geometric adjustment of the secondary radius.

This section derives the geometric compatibility constraint imposed by constant belt length. By expressing the total belt length in terms of the primary radius r_p, secondary radius r_s, and center-to-center distance C, we obtain an implicit relationship that couples the two pulley radii.

The derivation follows directly from the belt path geometry shown in Figure 13.

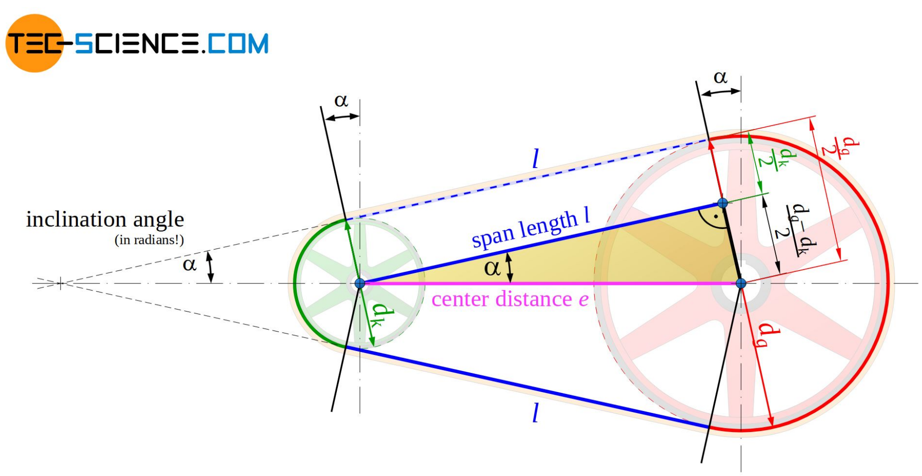

Figure 13: Belt geometry for two pulleys of radii r_p and r_s separated by center distance C. The belt wraps the pulleys with angles ϕ_p and ϕ_s and connects via two straight tangent spans ℓ.

Belt Length Decomposition The total belt length L_b is the sum of: (i) wrapped arc length on the primary pulley, (ii) wrapped arc length on the secondary pulley, and (iii) two straight tangent spans. Therefore,

$$L_b = r_p\phi_p + r_s\phi_s + 2\ell, \tag{7.3}$$

where ℓ is the length of a single straight tangent span segment.

Straight Span Length From the right-triangle geometry formed by the pulley centers and tangent points,

$$\ell = \sqrt{C^2 - (r_s - r_p)^2}. \tag{7.4}$$

Wrap Angles The wrap angles on the primary and secondary pulleys are

$$\phi_p = \pi - 2\lambda, \tag{7.5}$$
$$\phi_s = \pi + 2\lambda. \tag{7.6}$$

If $\lambda < 0$, these expressions automatically redistribute wrap between pulleys while preserving total belt length.

Tangency Angle (Signed Convention) Let λ denote the signed angle between the line of centers and the straight tangent belt segment (as defined in Figure 13). We define

$$\sin\lambda = \frac{r_s - r_p}{C}, \tag{7.7}$$

and equivalently,

$$\lambda = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.8}$$

Under this convention, λ may be negative when $r_p > r_s$. This signed definition ensures the wrap distribution automatically adjusts when the pulley ordering reverses.

Closed Form Belt-Length Expression Substituting Equation (7.4) and Equation (7.5)–Equation (7.6) into Equation (7.3) yields

$$\begin{aligned} L_b &= r_p(\pi - 2\lambda) + r_s(\pi + 2\lambda) + 2\sqrt{C^2 - (r_s - r_p)^2} \\ &= \pi(r_p + r_s) + 2\lambda(r_s - r_p) + 2\sqrt{C^2 - (r_s - r_p)^2}. \end{aligned} \tag{7.9}$$

Finally, substituting Equation (7.8) gives

$$L_b = \pi(r_p + r_s) + 2(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{7.10}$$

Root Form (Geometric Compatibility Constraint) Rearranging Equation (7.10) yields

Belt Length Constraint Root Function **(7.11)**

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b.$$

The belt-length constraint is therefore

$$f(r_p, r_s, C) = 0. \tag{7.12}$$

Remark (Transcendental Nature and Approximation)

Equation Equation (7.12) is transcendental due to the inverse-sine and square-root terms and therefore does not admit a closed-form solution.
In this work, the constraint is solved numerically using a bracketed root-finding method (e.g., bisection or Brent's method), which is deterministic and robust over the physically admissible interval.
It is common in simplified treatments to assume

$$\frac{r_s - r_p}{C} \approx 0,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this small-angle assumption, the belt-length equation reduces to a much simpler approximate form.
However, this approximation can introduce substantial geometric error when $\frac{r_s - r_p}{C}$ is not small. Appendix A quantifies this deviation and demonstrates that the approximation is not acceptable under realistic operating conditions.

7.5 Center-to-Center Distance Determination

In-order to make use of the belt-length constraint derived in Section 7.4, the center-to-center distance C must be determined.

The center-to-center distance C is a fixed geometric parameter of the assembled CVT system. Once the transmission is constructed, C does not vary during operation.

To determine C, we evaluate the belt-length constraint Equation (7.12) at the *known* initial configuration of the system.

Initial Geometric Configuration At startup, the CVT is assumed to be in its lowest ratio configuration:

- The primary pulley is at its minimum outer radius,

$$r_p = r_{p,\min}.$$

- The secondary pulley is at its maximum outer radius,

$$r_s = r_{s,\max}.$$

These values are determined by the physical geometry of the sheaves and represent the fully separated primary and fully closed secondary configuration.

Center Distance from Belt-Length Compatibility Substituting these known radii into Equation (7.12) yields

$$f(r_{p,\min}, r_{s,\max}, C) = 0. \tag{7.13}$$

This equation contains a single unknown, C, and is solved numerically to determine the fixed center-to-center distance consistent with the specified belt length L_b.

$$C = \underset{C}{\operatorname{root}}\; f(r_{p,\min}, r_{s,\max}, C)\,. \tag{7.14}$$

Remark (One-Time Geometric Solve)

The center distance C is determined once during model initialization. Thereafter, C is treated as a constant in all subsequent geometric and dynamic calculations.

7.6 Secondary Outer Radius as a Function of Shift

With the center-to-center distance C determined from Section 7.5, and the primary radius $r_p(s)$ defined by Equation (7.2), the secondary outer radius is obtained by enforcing the belt-length compatibility constraint.

Geometric Coupling At any shift position s, the primary radius $r_p(s)$ is known. The secondary radius r_s must therefore satisfy

$$f(r_p(s), r_s, C) = 0, \tag{7.15}$$

where $f(\cdot)$ is defined in Equation (7.11).

Numerical Determination of $r_s(s)$ Because Equation (7.15) is transcendental in r_s, the secondary radius is obtained numerically for each value of s:

Secondary Outer Radius **(7.16)**

$$r_{s,\text{outer}}(s) = \underset{r_s}{\operatorname{root}}\; f(r_p(s), r_s, C)\,.$$

Remark (Dynamic Root Solve)

Unlike the center-distance calculation in Section 7.5, which is performed once during initialization, this root solve is performed repeatedly as the shift coordinate s evolves during simulation. Because the physically admissible interval for r_s is bounded, a bracketed solver (e.g., Brent's method) converges rapidly.

With both $r_{p,\text{outer}}(s)$ and $r_{s,\text{outer}}(s)$ determined, all geometric quantities describing the belt path can now be expressed explicitly as functions of the shift coordinate s.

7.7 Wrap Angles as Functions of Shift

The wrap angles on the primary and secondary pulleys vary as the pulley radii change. Using the geometric relations from Section 7.4, the tangency angle becomes

$$\lambda(s) = \sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right). \tag{7.17}$$

Substitution into the wrap-angle definitions yields

$$\phi_p(s) = \pi - 2\sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right), \tag{7.18}$$

$$\phi_s(s) = \pi + 2\sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right). \tag{7.19}$$

7.8 Effective Torque Radius

All geometric relations derived thus far use the outer surface of the belt as the reference radius. This simplifies the belt-length formulation and pulley geometry.

Torque transmission, however, occurs through the belt cross-section. The appropriate moment arm for torque is therefore located approximately at the mid-height of the belt, as illustrated in Figure 14.

The effective torque radius for each pulley is defined as

$$r_{p,\text{eff}} = r_{p,\text{outer}} - \frac{h_b}{2}, \tag{7.20}$$

$$r_{s,\text{eff}} = r_{s,\text{outer}} - \frac{h_b}{2}. \tag{7.21}$$

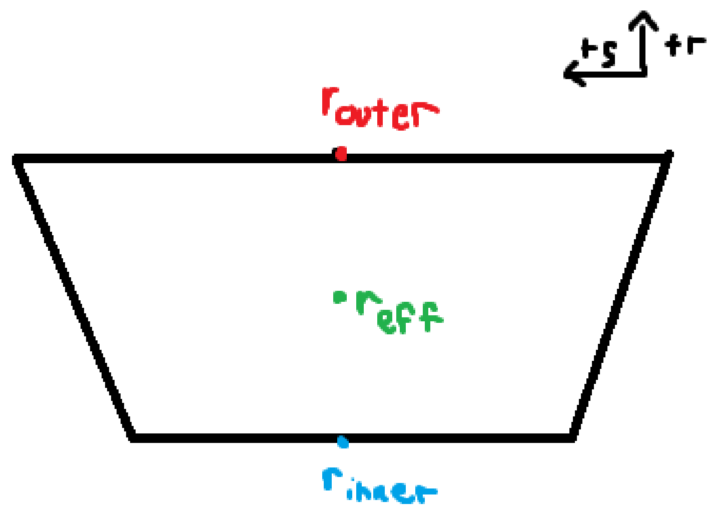

Figure 14: Belt cross-section showing the outer radius r_{outer}, inner radius r_{inner}, and effective torque radius r_{eff} located at the belt mid-height. The effective radius is used for all torque and power transmission calculations.

7.9 CVT Ratio Definition

The instantaneous transmission ratio is determined purely by pulley geometry. It is defined as the ratio of secondary to primary effective radii:

CVT Ratio **(7.22)**

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}.$$

Because both effective radii are functions of the shift coordinate s, the transmission ratio is likewise a function of s alone.

As the primary sheaves close, $r_{p,\text{eff}}(s)$ increases. By belt-length compatibility, the secondary radius decreases. Consequently, the ratio $R(s)$ decreases with increasing s.

Thus:

- Small s (minimum primary radius) corresponds to high ratio.
- Large s (maximum primary radius) corresponds to low ratio.

The CVT ratio is therefore a purely geometric quantity determined by the axial shift position.

7.10 Ratio Rate of Change

The CVT ratio is not only a function of shift position, but also changes in time whenever the sheaves are moving. This time variation matters because a changing ratio introduces additional kinematic coupling between the primary and secondary rotations. In other words, it is not sufficient to know the instantaneous ratio $R(s)$; the model also requires how quickly the ratio is changing.

This section derives an explicit expression for $\dot{R}$ in terms of the shift velocity $\dot{s}$ and geometric derivatives.

7.10.1 Start from the Ratio Definition

The ratio is defined by the effective radii as

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{7.23}$$

Because R depends on time only through $s(t)$, the chain rule gives

$$\dot{R} = \frac{dR}{ds}\dot{s}. \tag{7.24}$$

Thus, the objective is to compute dR/ds.

7.10.2 Differentiate R with Respect to s

Differentiating Equation (7.23) with respect to s (quotient rule) gives

$$\frac{dR}{ds} = \frac{d}{ds}\left(\frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}\right) = \frac{\left(\frac{dr_{s,\text{eff}}}{ds}\right) r_{p,\text{eff}} - r_{s,\text{eff}}\left(\frac{dr_{p,\text{eff}}}{ds}\right)}{r_{p,\text{eff}}^2}. \tag{7.25}$$

Rearranging into a form that exposes the required components:

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}}\frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\frac{dr_{p,\text{eff}}}{ds}. \tag{7.26}$$

At this point, the remaining task is to express the two derivatives $dr_{p,\text{eff}}/ds$ and $dr_{s,\text{eff}}/ds$ in computable form.

7.10.3 Eliminating dr_s/ds Using the Belt-Length Constraint

The secondary radius $r_s(s)$ is defined implicitly through the belt-length constraint and does not admit a closed-form expression in terms of r_p. Direct differentiation of $r_s(s)$ is therefore not practical.

Instead, we differentiate the belt-length compatibility constraint derived in Section 7.4, which is written as

$$f(r_p, r_s, C) = 0, \tag{7.27}$$

with $f(\cdot)$ defined explicitly in Equation (7.11) and C constant. Since $r_p = r_p(s)$ and $r_s = r_s(s)$, differentiating Equation (7.27) with respect to s gives (multivariable chain rule)

$$\frac{\partial f}{\partial r_p}\frac{dr_p}{ds} + \frac{\partial f}{\partial r_s}\frac{dr_s}{ds} = 0. \tag{7.28}$$

Solving Equation (7.28) for dr_s/ds yields

$$\frac{dr_s}{ds} = -\frac{\dfrac{\partial f}{\partial r_p}}{\dfrac{\partial f}{\partial r_s}} \frac{dr_p}{ds}. \tag{7.29}$$

7.10.4 Single Expression for dR/ds in Terms of Remaining Unknowns

Substituting Equation (7.29) into Equation (7.26) eliminates dr_s/ds and gives a single expression for dR/ds in terms of the belt constraint partial derivatives and the primary geometric derivative:

$$\frac{dR}{ds} = -\frac{1}{r_{p,\text{eff}}} \frac{\dfrac{\partial f}{\partial r_p}}{\dfrac{\partial f}{\partial r_s}} \frac{dr_p}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \frac{dr_{p,\text{eff}}}{ds}. \tag{7.30}$$

Equation Equation (7.30) makes the remaining requirements explicit. To evaluate dR/ds (and therefore $\dot{R}$ via Equation (7.24)), three quantities are required:

1. $\dfrac{\partial f}{\partial r_p}$,
2. $\dfrac{\partial f}{\partial r_s}$,
3. $\dfrac{dr_p}{ds}$ (and the associated $\dfrac{dr_{p,\text{eff}}}{ds}$).

The following subsections derive these three quantities for the specific belt-length function f defined in Equation (7.11) and for the primary geometry defined in Equation (7.2).

7.10.5 Partial Derivatives of the Belt Constraint

Recall the belt-length constraint written in root form as

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b = 0. \tag{7.31}$$

We now compute the two partial derivatives required in Equation (7.29), differentiating term-by-term and applying the product rule directly to the inverse-sine term.

Product rule for the inverse-sine factor Consider the factor

$$(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right).$$

Differentiating with respect to r_p (product rule) gives

$$\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \underbrace{\frac{\partial}{\partial r_p}(r_s - r_p)}_{(1)}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\underbrace{\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right)}_{(2)}. \tag{7.32}$$

We therefore compute the two sub-derivatives.

Derivative (1)

$$\frac{\partial}{\partial r_p}(r_s - r_p) = -1. \tag{7.33}$$

Derivative (2) Using the chain rule,

$$\frac{d}{dx}\sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}},$$

let

$$u = \frac{r_s - r_p}{C}.$$

Then

$$\frac{\partial u}{\partial r_p} = -\frac{1}{C}.$$

Therefore,

$$\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) = \frac{1}{\sqrt{1-u^2}}\left(-\frac{1}{C}\right). \tag{7.34}$$

Now simplify the denominator:

$$\sqrt{1-u^2} = \sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2} = \frac{\sqrt{C^2 - (r_s - r_p)^2}}{C}.$$

Substituting into Equation (7.34) yields

$$\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) = -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.35}$$

Substitute into the product rule Substituting Equation (7.33) and Equation (7.35) into Equation (7.32) gives

$$\begin{aligned}\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] &= (-1)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\left(-\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}\right) \\ &= -\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \end{aligned} \tag{7.36}$$

Derivative of the square-root term For the straight-span contribution,

$$2\sqrt{C^2 - (r_s - r_p)^2},$$

we apply the chain rule:

$$\frac{\partial}{\partial r_p}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.37}$$

Assemble $\partial f/\partial r_p$ Differentiating Equation (7.31) term-by-term gives

$$\frac{\partial f}{\partial r_p} = \pi + 2\left(-\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) + \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}$$
$$= \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.38}$$

The square-root contributions cancel exactly.

Derivative with respect to r_s The same procedure yields

$$\frac{\partial}{\partial r_s}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}, \tag{7.39}$$

and

$$\frac{\partial}{\partial r_s}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = -\frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.40}$$

Assembling terms gives

$$\frac{\partial f}{\partial r_s} = \pi + 2\left(\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) - \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}$$
$$= \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.41}$$

Summary of Belt Constraint Partial Derivatives From the derivations above, the partial derivatives of the belt-length constraint Equation (7.31) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{7.42}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.43}$$

These expressions are globally valid under the geometric definition of λ given in Equation (7.8) and require no piecewise case distinction.

7.10.6 Primary Radius Derivative

From the piecewise primary radius model Equation (7.2), the primary outer radius satisfies

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\min}, & 0 \le s < s_{\text{dz}}, \\ r_{p,\min} + \dfrac{s - s_{\text{dz}}}{2\tan\beta}, & s_{\text{dz}} \le s \le s_{\max}. \end{cases}$$

Differentiating with respect to s gives

$$\frac{dr_p}{ds} = \frac{dr_{p,\text{outer}}}{ds} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\ \dfrac{1}{2\tan\beta}, & s_{\text{dz}} \le s \le s_{\max}. \end{cases} \tag{7.44}$$

Because $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ and h_b is constant, the same derivative applies:

$$\frac{dr_{p,\text{eff}}}{ds} = \frac{dr_p}{ds}. \tag{7.45}$$

Remark (At the Travel Limit)

If the model clamps s at the physical limit $s_{\max}$, then $r_{p,\text{outer}}$ (and thus $r_{p,\text{eff}}$) is constant beyond this point and the derivative is zero for $s > s_{\max}$. In this document we restrict to the admissible domain $0 \le s \le s_{\max}$.

7.10.7 Final Expression for dR/ds and $\dot{R}$

Equation (7.30) expresses dR/ds in terms of three quantities: (i) the belt-constraint partial derivatives $\partial f/\partial r_p$ and $\partial f/\partial r_s$, and (ii) the primary geometric derivatives dr_p/ds and $dr_{p,\text{eff}}/ds$.

From Section 7.10.5, the partial derivatives of the belt-length constraint Equation (7.11) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{7.46}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.47}$$

From Section 7.10.6, the primary outer-radius derivative is Equation (7.44), and since $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ with h_b constant, Equation (7.45) gives $dr_{p,\text{eff}}/ds = dr_p/ds$.

Substituting Equation (7.46)–Equation (7.47) and Equation (7.45) into Equation (7.30) yields

$$\frac{dR}{ds} = \left[-\frac{1}{r_{p,\text{eff}}} \frac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \right] \frac{dr_p}{ds}. \tag{7.48}$$

Finally, substituting the piecewise primary derivative Equation (7.44) gives the fully expanded result

$$\frac{dR}{ds} = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ \dfrac{1}{2\tan\beta}\left[-\dfrac{1}{r_{p,\mathrm{eff}}}\dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} - \dfrac{r_{s,\mathrm{eff}}}{r_{p,\mathrm{eff}}^2}\right], & s_{\mathrm{dz}} \le s \le s_{\max}. \end{cases} \tag{7.49}$$

Using the chain rule Equation (7.24), $\dot{R} = (dR/ds)\dot{s}$. Substituting Equation (7.49) and rearranging gives

Rate of Change of Transmission Ratio **(7.50)**

$$\dot{R} = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ -\dfrac{\dot{s}}{2\tan\beta\, r_{p,\mathrm{eff}}}\left[\dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} + \dfrac{r_{s,\mathrm{eff}}}{r_{p,\mathrm{eff}}}\right], & s_{\mathrm{dz}} \le s \le s_{\max}. \end{cases}$$

Summary Equations Equation (7.49) and Equation (7.50) give the geometric sensitivity of the ratio to axial motion and its time rate of change. The deadzone behavior is explicit: when $0 \le s < s_{\mathrm{dz}}$, the ratio is constant and $\dot{R} = 0$. In the active region, $s_{\mathrm{dz}} \le s \le s_{\max}$, the ratio evolves according to geometry and the shift rate $\dot{s}$.

7.11 Section Summary and Forward Connection

This section established the complete *geometric foundation* of the CVT model. The development was intentionally kinematic: the objective was to derive a closed, computable mapping from axial shift to pulley geometry and transmission ratio, without introducing force balance, slip behavior, or torque relations.

At a high level, the geometry defines the chain

$$s(t) \longrightarrow r_{p,\mathrm{eff}}(s),\ r_{s,\mathrm{eff}}(s) \longrightarrow R(s) \longrightarrow \dot{R}(s, \dot{s}),$$

with each intermediate relationship derived explicitly from pulley and belt geometry.

Key Results Established The following geometric components are now defined and ready for use in subsequent modeling:

1. **Primary radius law.** Equation (7.2) provides the piecewise relationship between axial shift s and primary outer radius, including deadzone behavior and cone-angle dependence.
2. **Belt-length compatibility constraint.** Equation (7.12) enforces constant belt length and defines the implicit, nonlinear coupling between r_p, r_s, and the center distance C.

3. **Center distance determination.** Section 7.5 fixes the assembled center-to-center distance C by evaluating the belt constraint at a known geometric configuration. This solve is performed once during initialization.
4. **Secondary radius as a function of shift.** Section 7.6 defines $r_s(s)$ implicitly by solving the belt constraint at each shift position, producing the required redistribution of belt radius between pulleys.
5. **Wrap angles as functions of shift.** Section 7.7 defines the wrap angles $\phi_p(s)$ and $\phi_s(s)$ follow directly from the tangency geometry and are available for later contact, work, and belt-distribution calculations.
6. **Effective torque radii.** Section 7.8 corrects outer radii to effective radii using belt thickness, ensuring subsequent torque relations use an appropriate moment arm.
7. **Ratio and ratio rate.** Equation (7.22) defines the geometric ratio $R(s)$, and Section 7.10 provides $\dfrac{dR}{ds}$ and $\dot{R}$ via Equation (7.49) and Equation (7.50), including the deadzone condition where R is constant.

Structural Role in the Full Model These results form the geometric backbone of the CVT simulation. All subsequent sections treat the relationships derived here as fixed mappings. In particular, later torque and force models will use $r_{p,\text{eff}}(s)$ and $r_{s,\text{eff}}(s)$ as inputs, and shifting kinematics will use $R(s)$ and $\dot{R}(s, \dot{s})$ to relate primary and secondary motion during ratio change.

Several key properties are now clear: (i) the geometric coupling is inherently nonlinear (through the belt-length constraint), (ii) the deadzone is explicitly represented, and (iii) within the admissible travel range the mappings are differentiable and suitable for continuous-time simulation.

Transition to Dynamics With the geometry fully specified, the next objective is to determine how the shift coordinate evolves in time. Subsequent sections introduce axial force generation (from pulley mechanisms and belt interaction) to determine $\dot{s}$ and $\ddot{s}$. Once $\dot{s}$ is known, the ratio-rate expression Equation (7.50) provides the corresponding ratio evolution and the required kinematic coupling between rotating components during shifting.

8 Pulley Axial Force Models

The geometric relations developed in the previous section define how the transmission ratio varies with axial shift. What remains is to determine *why* the shift coordinate $s(t)$ changes in time.

Axial motion of the sheaves is driven by clamping forces generated within the primary and secondary assemblies. On the primary side, centrifugal flyweights and springs produce an inward axial force that tends to close the sheaves and increase radius. On the secondary side, torque reaction and spring preload generate an opposing axial force.

The evolution of the shift coordinate is governed by the net axial force acting on the movable sheave assembly. Applying Newton's second law in the axial direction (Theory 1 (Newton's Second Law (Translation))) gives

$$m\,\ddot{s} = F_{p,\text{ax}} - F_{s,\text{ax}}, \tag{8.1}$$

where $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ denote the axial forces generated by the primary and secondary mechanisms, respectively, and m is the effective axial inertia of the moving components.

The purpose of this section is to derive expressions for $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ as functions of the current system state. Once these force models are established, the shift acceleration $\ddot{s}$ follows directly, completing the dynamic description of axial motion.

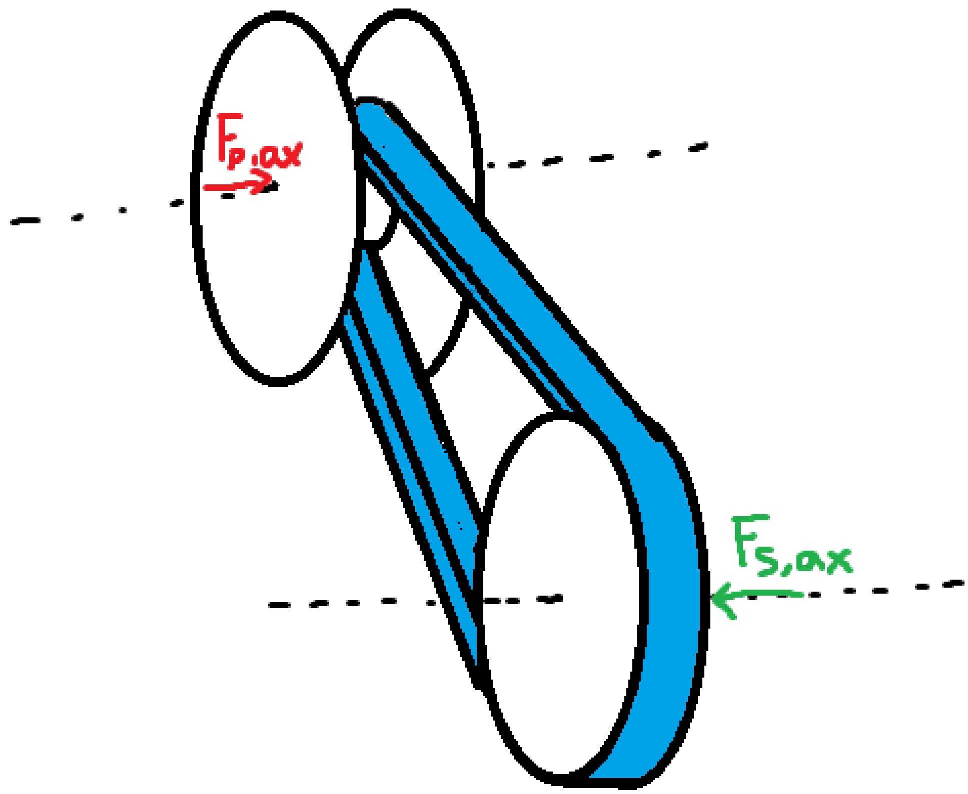

Figure 15: Axial force balance governing sheave motion. Primary and secondary mechanisms generate opposing axial forces. The net force determines the shift acceleration $\ddot{s}$.

Although the specific actuator mechanisms modeled here correspond to a flyweight-driven primary and a torque-reactive helix secondary, the formulation is modular: any actuator model may be substituted provided it returns the net axial force as a function of state.

8.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined contact surfaces. As the shift coordinate s evolves, mechanical elements slide along these surfaces, converting radial or circumferential forces into axial clamping force.

To keep the formulation general and modular, the ramp geometries are parameterized directly as functions of the axial shift coordinate s. Any actuator profile may therefore be substituted provided it supplies the appropriate geometric mapping and derivative.

Primary Ramp Geometry (Axial-to-Radial Mapping) On the primary pulley, centrifugal force acting on the flyweights is redirected into axial closing force through a ramp surface (see Figure 16). As the movable sheave translates axially, the flyweight slides along this ramp.

We represent the ramp by an axial-to-radial profile

$$r_f = r_f(s), \tag{8.2}$$

where $r_f(s)$ is the effective flyweight radius.

An incremental axial displacement ds produces a radial displacement dr_f. The local slope of the ramp is therefore dr_f/ds. The ramp angle is defined with respect to the axial direction as

$$\tan\alpha_p(s) = \frac{dr_f}{ds}, \qquad \alpha_p(s) = \tan^{-1}\left(\frac{dr_f}{ds}\right). \tag{8.3}$$

This definition follows directly from the geometry of the sliding motion: a steeper ramp (larger dr_f/ds) produces greater radial change per unit axial travel and therefore greater axial gain when radial force is redirected through the surface.

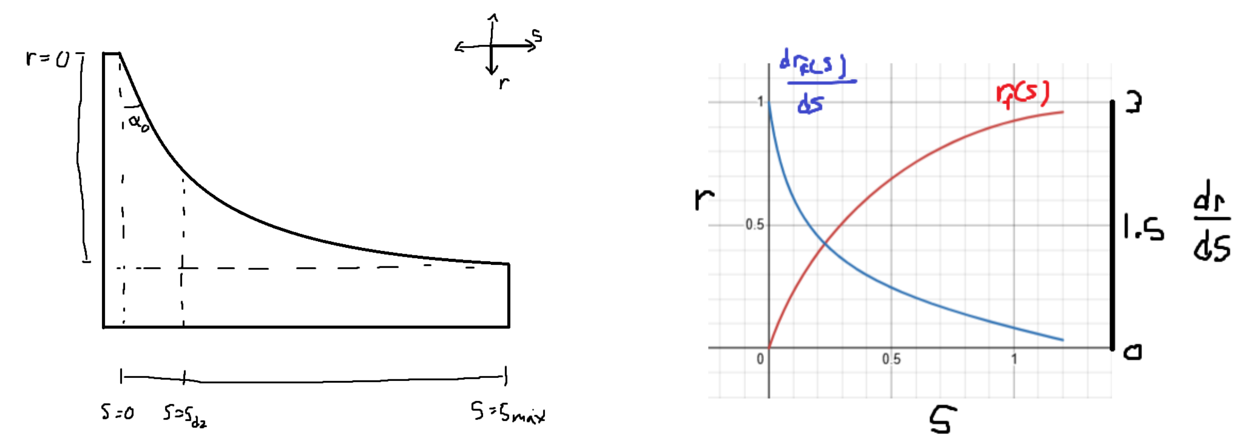

Figure 16: Primary ramp geometry. **Left:** As axial displacement s increases, the flyweight slides along the ramp, producing radial change dr_f. The instantaneous ramp angle satisfies $\tan\alpha_p = dr_f/ds$. **Right:** Example profiles of $r_f(s)$ and its slope dr_f/ds.

Primary Admissibility Conditions In order to be a valid primary ramp profile $r_f(s)$, it must satisfy:

1. $r_f(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{dr_f}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{dr_f}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the ramp angle $\alpha_p(s)$ is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below $\pi/2$ and the axial gain $\tan\alpha_p$ remains finite.

Secondary Helix Geometry (Axial-to-Rotation Mapping) On the secondary pulley, axial travel produces relative rotation of a helical ramp mechanism, loading a torsional spring and generating torque-reactive clamping (see Figure 17).

We parameterize this kinematically as

$$\theta_s = \theta_s(s), \tag{8.4}$$

where $\theta_s(s)$ is the helix-induced rotation.

An incremental rotation $d\theta_s$ corresponds to a circumferential displacement $r_h d\theta_s$ along a cylindrical surface of effective radius r_h. The helix angle is defined from the local slope of the ramp surface:

$$\tan\alpha_s = \frac{\text{axial rise}}{\text{circumferential run}} = \frac{ds}{r_h d\theta_s}.$$

Rewriting in differential form gives

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}. \tag{8.5}$$

Thus, a shallower helix (smaller α_s) produces greater rotation per unit axial travel, while a steeper helix produces less rotation. This inverse relationship arises naturally from the geometry of motion on a cylindrical surface.

Remark (Contrast with Primary Ramp)

The primary ramp behaves like a simple wedge: radial displacement is directly proportional to axial travel, so the ramp angle is naturally defined by $\tan\alpha_p = dr_f/ds$.
The secondary mechanism instead behaves like a power screw. Axial motion is produced through rotation around a cylindrical surface, so the relevant geometric ratio is how much rotation occurs per unit axial travel. This leads to the slope definition $ds/(r_h d\theta_s)$ and the inverse relationship in Equation (8.5).
In short, the primary redirects force through a wedge, while the secondary converts torque through a screw-like helix. The difference is purely geometric and reflects the distinct mechanisms by which axial motion is generated.

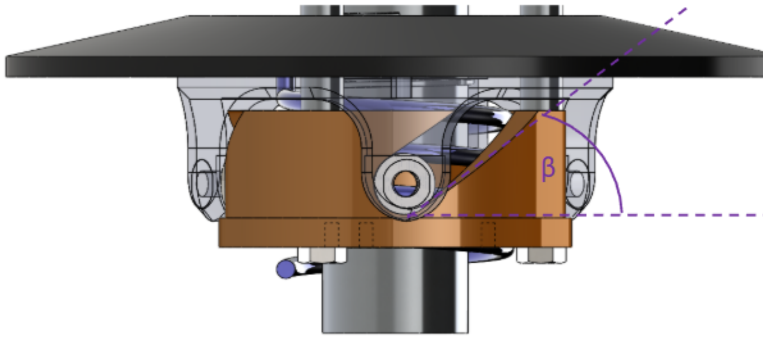

Figure 17: Secondary helix geometry. Axial travel s induces rotation $\theta_s(s)$ around a cylindrical surface of radius r_h. The instantaneous helix angle satisfies $\tan\alpha_s = ds/(r_h d\theta_s)$.

Secondary Admissibility Conditions The helix rotation mapping $\theta_s(s)$ must satisfy:

1. $\theta_s(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{d\theta_s}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{d\theta_s}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases rotation. Condition (3) ensures the effective helix angle remains strictly below $\pi/2$ and the geometric gain remains finite.

Remark (Piecewise Profiles in Practice)

The C^1 requirement above is sufficient for all derivations in this document. In practice, piecewise C^1 profiles (continuous everywhere, continuously differentiable except at finitely many junction points where left and right derivatives exist) are also admissible for numerical simulation. At junction points, one-sided derivatives provide the required slope values, and the resulting force discontinuities pose no difficulty for standard ODE solvers. However, large slope discontinuities can introduce local stiffness into the system: the abrupt change in force gradients may require adaptive solvers to reduce the timestep significantly near the junction, degrading computational efficiency. In extreme cases, explicit integrators may struggle to maintain stability without impractically small step sizes, and implicit or stiff-aware solvers may be preferable.

Derivations for common ramp and helix profiles are provided in Appendix B.

8.2 Primary Axial Force Model

The primary axial clamping force is generated by the interaction between centrifugal loading of the flyweights and the restoring force of the primary compression spring.

As shown in Figure 18, each flyweight of mass m_f rotates with angular speed ω_p and experiences an outward radial centrifugal force. This radial force acts along the ramp profile $r_f(s)$ and is redirected into an axial closing force through the ramp geometry described in Section 8.1.

Opposing this motion is the primary compression spring, characterized by stiffness k_p and preload $x_{p,0}$. As the sheaves close (s increases), the spring compresses and produces a restoring axial force resisting further closure.

The net axial force on the movable primary sheave is therefore the axial component of the ramp-redirected centrifugal force minus the spring force. This net force determines the axial acceleration $\ddot{s}$ of the primary assembly.

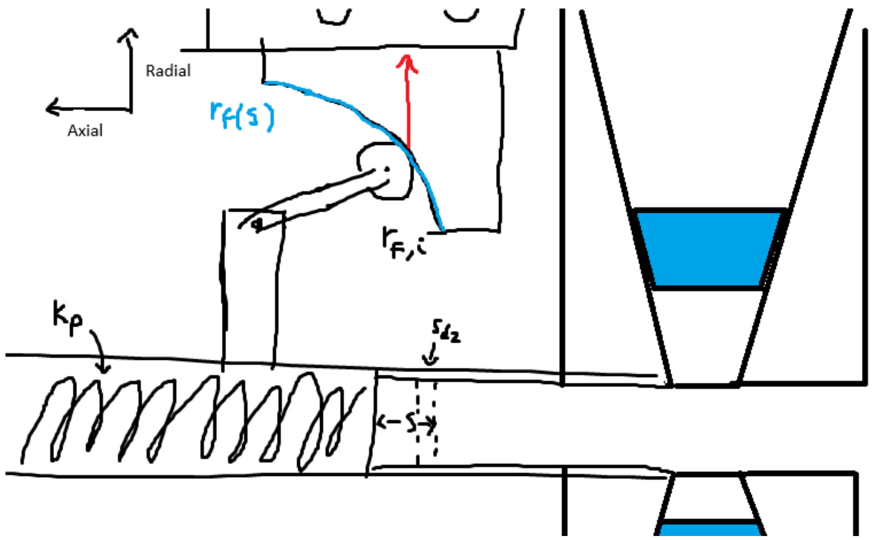

Figure 18: Primary CVT assembly showing flyweight mass m_f, ramp profile $r_f(s)$, compression spring with stiffness k_p and preload $x_{p,0}$, and axial shift coordinate s.

The state variable governing primary rotation is ω_p.

Centrifugal Loading of Flyweights From Theory 5 (Centrifugal Force),

$$F_c = m\omega^2 r.$$

Let

- m_f denote the flyweight mass,
- $r_{f,0}$ denote the initial flyweight radius at $s = 0$,
- $r_f(s)$ denote the incremental radial change produced by the ramp.

The total effective flyweight radius is therefore

$$r_{f,\text{tot}}(s) = r_{f,0} + r_f(s).$$

The radial centrifugal force is

$$F_{p,\text{rad}}(s,\omega_p) = m_f\omega_p^2\big(r_{f,0} + r_f(s)\big). \tag{8.6}$$

Radial-to-Axial Projection The ramp redirects radial force into axial force. As shown in Figure 19, the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan\alpha_p(s). \tag{8.7}$$

From Equation (8.3), the ramp angle satisfies

$$\tan\alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting gives

$$F_{p,\text{cent}}(s,\omega_p) = m_f\omega_p^2\big(r_{f,0} + r_f(s)\big)\frac{dr_f}{ds}. \tag{8.8}$$

Thus the axial gain of centrifugal force is governed by the local ramp slope, while the centrifugal magnitude depends on the total flyweight radius.

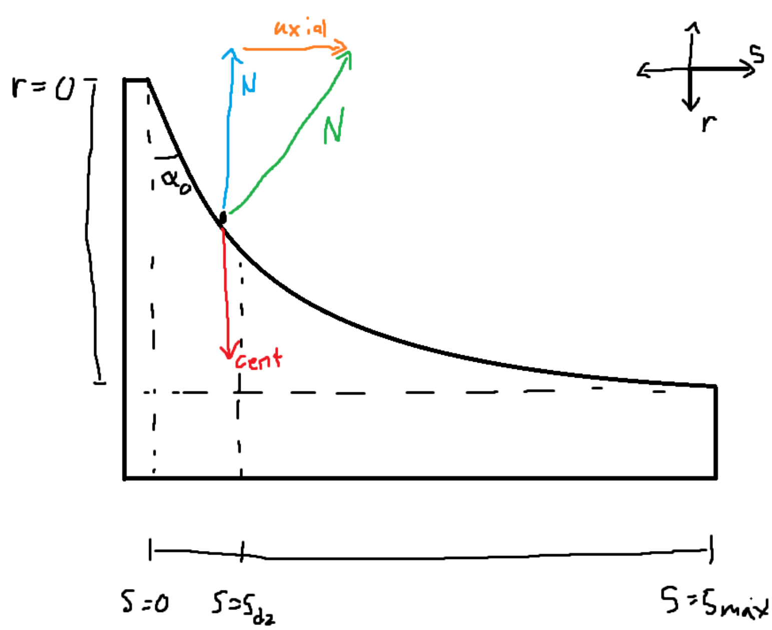

Figure 19: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of $\tan\alpha_p$, which equals dr_f/ds by definition of the ramp slope.

Primary Compression Spring From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let k_p denote primary stiffness and $x_{p,0}$ preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \tag{8.9}$$

Net Primary Axial Force The net axial force generated by the primary mechanism is the difference between the centrifugal axial component and the spring force:

Primary Axial Force **(8.10)**

$$F_{p,\text{ax}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds} - k_p(x_{p,0} + s).$$

8.3 Secondary Axial Force Model

The secondary pulley is *torque-reactive*: its axial clamping force is generated in response to torque transmitted through the CVT. Unlike the primary, which produces clamping based on rotational speed, the secondary produces clamping primarily when torque flows through it.

The mechanism consists of a helical ramp between the two sheaves and a single spring located between them. This spring serves two roles:

- It acts axially in compression as the sheaves separate.
- It resists relative rotation torsionally as the helix turns.

Thus, axial motion and rotational motion of the helix both load the same spring.

Figure 20: Secondary CVT assembly showing the helix mechanism and spring. Torque τ_s transmitted through the belt produces tangential force at the helix ramps, which is redirected into axial clamping force.

Physical Mechanism (Screw Analogy) The secondary helix behaves mechanically like a screw.

If you try to spin a screw in free space, nothing happens axially. If you spin a screw into a block that is not restrained, the block simply rotates with the screw—again producing no axial force. Axial force only develops when there is resistance to rotation on one side while torque is applied on the other.

In other words, a screw converts *reacted torque* into axial force.

The secondary helix operates on the same principle. The belt applies a driving torque to the secondary pulley. If the driven shaft were free, the entire assembly would rotate without developing tangential reaction force at the helix. However, when the driven load resists rotation, torque is transmitted through the helix ramps.

Those ramps are inclined surfaces—just like screw threads. As torque attempts to rotate one helix face relative to the other, contact forces develop along the inclined surfaces. Because the surfaces are angled, this tangential reaction force has an axial component.

The result is axial clamping force.

The key distinction from a conventional screw is direction: in a normal screw, torque pulls two components together. In the CVT helix, the orientation of the ramp is such that torque causes the movable sheave to be pushed outward, increasing belt clamping.

Thus, the secondary does not clamp because it spins. It clamps because torque is being transmitted and resisted.

Definition of Transmitted Torque Let τ_s denote the torque transmitted through the secondary shaft. This is the torque carried from the belt, through the helix mechanism, and into the driven load.

For the present derivation, τ_s is treated as a known input. Its value will be computed later in Section 9.

Torsional Spring Torque As the helix rotates relative to the opposing sheave, the spring is twisted torsionally.

From Theory 7 (Hooke's Law (Torsional)),

$$\tau = k\Delta\theta.$$

Let $k_{s,\theta}$ denote torsional stiffness and $\theta_{s,0}$ preload.

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big). \tag{8.11}$$

This torsional component of the same spring contributes to helix loading even when no torque is transmitted ($\tau_s = 0$), providing baseline clamping.

Total Torque Applied to Helix

$$\tau_{s,\text{total}} = \tau_s + \tau_{s,\text{spring}}(s). \tag{8.12}$$

Both contributions act in the same rotational direction at the helix. The transmitted torque loads the helix through belt force, and the torsional spring resists relative rotation. Both therefore generate axial clamping through the same geometry.

Torque-to-Axial Conversion Through the Helix From Theory 3 (Torque Definition),

$$\tau = F_t r_h \quad \Rightarrow \quad F_t = \frac{\tau_{s,\text{total}}}{r_h}.$$

The tangential force F_t acts along two symmetric helix ramp faces. Each ramp carries half of the total tangential load, introducing the factor of $1/2$ in the axial projection below.

From helix geometry (see Figure 21),

$$F_{s,\text{helix}} = \frac{F_t}{2\tan\alpha_s(s)}. \tag{8.13}$$

Substituting F_t and using

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}$$

gives

$$F_{s,\text{helix}}(s,\tau_s) = \frac{\tau_{s,\text{total}}}{2}\frac{d\theta_s}{ds}. \tag{8.14}$$

Notably, the helix radius r_h cancels: torque is converted into axial force according to the geometric rotation-per-travel relationship $d\theta_s/ds$, consistent with screw mechanics.

TODO: Remark on less friction at bigger radius if being considered

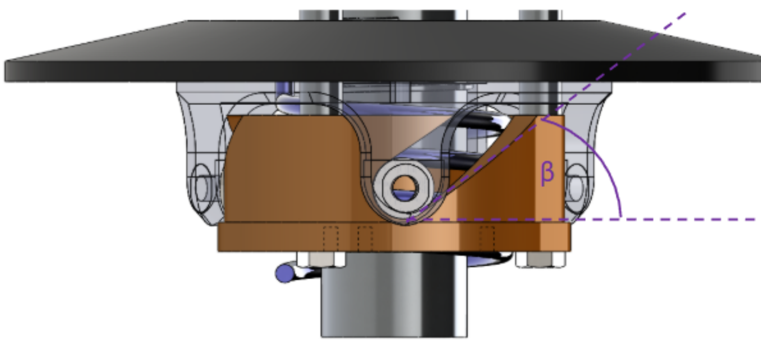

Figure 21: Helix free-body concept. Tangential force from shaft torque is redirected into axial force along the inclined ramps. Two symmetric ramp faces share the tangential load, producing the 1/2 factor.

Secondary Compression Spring The same spring also acts axially between the sheaves. At $s = 0$, the spring is pre-compressed by an amount $x_{s,0}$, producing a baseline separating force that assists upshift.

As the sheaves move apart (increasing s), the spring extends and its compression decreases.

From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let $k_{s,x}$ denote axial stiffness. Because increasing s reduces compression, the effective compression is

$$\Delta x = x_{s,0} - s.$$

Thus,

$$F_{s,\text{comp}}(s) = k_{s,x}(x_{s,0} - s). \tag{8.15}$$

This axial spring force assists upshift at small s, but weakens progressively as the sheaves separate.

Net Secondary Axial Force The total secondary axial force is the sum of:

- Helix-generated torque-reactive clamping,
- Torsional spring contribution (through the helix),
- Direct axial compression spring force.

Secondary Axial Force **(8.16)**

$$F_{s,\text{ax}}(s, \tau_s) = \underbrace{\frac{\tau_s + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2}\frac{d\theta_s}{ds}}_{\text{torque-reactive helix (including torsional spring)}} + \underbrace{k_{s,x}(x_{s,0} - s)}_{\text{axial spring (weakening with shift)}}.$$

8.4 Belt Centrifugal Force on the Wrap

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Centrifugal Force Formulation From Theory 5 (Centrifugal Force), a mass m rotating at angular velocity ω at radius r experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around a pulley, we require:

1. The mass of belt material in contact with the pulley.
2. The radius at which this mass rotates.
3. The angular velocity of the belt.

Each of these quantities is derived below.

Belt Cross-Section Geometry The belt cross-section is approximated as a trapezoid (see Figure 22) with:

- radial thickness h_b,
- outer width b_{out} (at the belt's outer surface),
- inner width b_{in} (at the belt's inner surface).

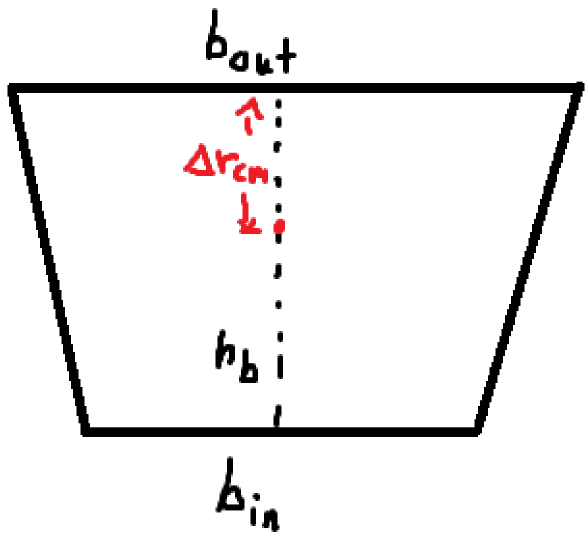

Figure 22: Trapezoidal belt cross-section. The outer width b_{out} corresponds to the surface away from the pulley axis; the inner width b_{in} corresponds to the surface closer to the axis. The radial thickness is h_b and the local width varies linearly between b_{out} and b_{in}, producing a δr_{cm} offset of the mass centroid from the geometric center.

The cross-sectional area of the trapezoid is

$$A_b = \frac{h_b}{2}\left(b_{\text{out}} + b_{\text{in}}\right). \tag{8.17}$$

Let ρ_b denote the belt material density (kg/m^3). The mass of a differential belt segment of arc length $\mathrm{d}\ell$ is therefore

$$\mathrm{d}m = \rho_b A_b \, \mathrm{d}\ell. \tag{8.18}$$

Belt Mass Centroid Centrifugal force acts at the center of mass of the rotating belt element, not at the outer or inner surface. Because the belt cross-section is trapezoidal (not rectangular), the centroid is offset from the geometric center.

Let y denote radial distance measured inward from the outer surface, so that $y = 0$ at the outer surface and $y = h_b$ at the inner surface. For a trapezoid with linear sidewalls, the local width varies linearly with y:

$$b(y) = b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y. \tag{8.19}$$

The centroid location Δr_{cm}, measured inward from the outer surface, is computed from the definition of the centroid of an area:

$$\Delta r_{\text{cm}} = \frac{1}{A_b} \int_0^{h_b} y \, b(y) \, \mathrm{d}y. \tag{8.20}$$

Substituting Equation (8.19) into this integral:

$$
\begin{aligned}
\int_0^{h_b} y\, b(y)\, \mathrm{d}y &= \int_0^{h_b} y \left[b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y\right] \mathrm{d}y \\
&= \int_0^{h_b} b_{\text{out}}\, y\, \mathrm{d}y \;+\; \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \int_0^{h_b} y^2\, \mathrm{d}y \\
&= b_{\text{out}} \left[\frac{y^2}{2}\right]_0^{h_b} \;+\; \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \left[\frac{y^3}{3}\right]_0^{h_b} \\
&= b_{\text{out}} \cdot \frac{h_b^2}{2} \;+\; \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \cdot \frac{h_b^3}{3} \\
&= \frac{h_b^2}{2} b_{\text{out}} \;+\; \frac{h_b^2}{3} (b_{\text{in}} - b_{\text{out}}) \\
&= \frac{h_b^2}{6} (3b_{\text{out}} + 2b_{\text{in}} - 2b_{\text{out}}) \\
&= \frac{h_b^2}{6} (b_{\text{out}} + 2b_{\text{in}})\,.
\end{aligned} \tag{8.21}
$$

Dividing by the area $A_b = \frac{h_b}{2}(b_{\text{out}} + b_{\text{in}})$ from Equation (8.17):

$$
\Delta r_{\text{cm}} = \frac{\frac{h_b^2}{6} (b_{\text{out}} + 2b_{\text{in}})}{\frac{h_b}{2} (b_{\text{out}} + b_{\text{in}})} = h_b \cdot \frac{b_{\text{out}} + 2b_{\text{in}}}{3\,(b_{\text{out}} + b_{\text{in}})}. \tag{8.22}
$$

The belt mass-centroid radius at any shift position is therefore

$$
r_{\text{cm}}(s) = r_{\text{out}}(s) - \Delta r_{\text{cm}}, \tag{8.23}
$$

where $r_{\text{out}}(s)$ is the outer belt radius at the current shift (equal to the pulley outer radius at that position).

Distributed Centrifugal Load Along the Wrap Consider a differential belt element along the wrapped arc. Let θ denote the angular coordinate measured from the center of the wrap, so that the wrapped arc spans

$$
\theta \in \left[-\frac{\phi}{2}, \frac{\phi}{2}\right],
$$

where ϕ is the total wrap angle (see Figure 23).

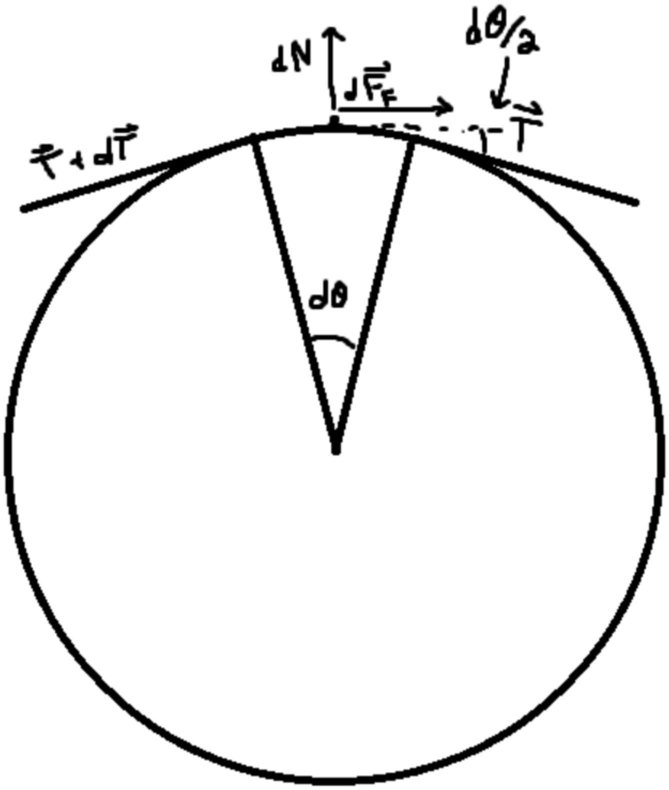

Figure 23: Belt wrapped around a pulley. A differential element at angle θ has arc length $\mathrm{d}\ell = r_{\mathrm{cm}}\ \mathrm{d}\theta$ and experiences radially outward centrifugal force.

A differential belt element has arc length

$$\mathrm{d}\ell = r_{\mathrm{cm}}\ \mathrm{d}\theta.$$

From Equation (8.18), the corresponding differential mass is

$$\mathrm{d}m = \rho_b A_b r_{\mathrm{cm}}\ \mathrm{d}\theta. \tag{8.24}$$

From Theory 5 (Centrifugal Force), the centrifugal force on this element is

$$\mathrm{d}F_c = \mathrm{d}m\,\omega^2 r_{\mathrm{cm}} = \rho_b A_b\,\omega^2 r_{\mathrm{cm}}^2\ \mathrm{d}\theta. \tag{8.25}$$

This force acts radially outward at each point along the wrap.

Total Radial Force Over the Wrap Integrating Equation (8.25) over the entire wrap angle yields the total radially-directed centrifugal force:

$$
\begin{aligned}
F_{c,\mathrm{rad}} &= \int_{-\phi/2}^{\phi/2} \rho_b A_b\, \omega^2 r_{\mathrm{cm}}^2 \,\mathrm{d}\theta \\
&= \rho_b A_b\, \omega^2 r_{\mathrm{cm}}^2 \int_{-\phi/2}^{\phi/2} \mathrm{d}\theta \\
&= \rho_b A_b\, \omega^2 r_{\mathrm{cm}}^2\, \phi.
\end{aligned} \tag{8.26}
$$

Radial-to-Axial Conversion Through the Sheave Cone The radial centrifugal force acts outward against the conical sheave faces. Because the sheaves are inclined at cone half-angle β, radial loading cannot occur without simultaneous axial motion of the sheaves. Thus, a purely radial force applied to the belt generates an axial reaction component that contributes to clamping.

From the pulley geometry derived previously (see Equation (7.1)), the relationship between radial and axial displacement in the active shift region is

$$
\frac{\mathrm{d}r}{\mathrm{d}s} = \frac{1}{2\tan\beta}.
$$

This relation describes how radial motion of the belt corresponds to axial motion of the sheaves. To convert force components, we enforce virtual work consistency between radial and axial motion.

For an infinitesimal displacement,

$$
\delta W = F_{\mathrm{rad}}\, \mathrm{d}r = F_{\mathrm{ax}}\, \mathrm{d}s.
$$

Substituting $\mathrm{d}r = \dfrac{1}{2\tan\beta}\,\mathrm{d}s$ gives

$$
F_{\mathrm{rad}} \frac{1}{2\tan\beta}\,\mathrm{d}s = F_{\mathrm{ax}}\,\mathrm{d}s.
$$

Canceling $\mathrm{d}s$ yields

$$
F_{\mathrm{ax}} = \frac{F_{\mathrm{rad}}}{2\tan\beta}.
$$

Thus, the total axial clamping contribution from belt centrifugal force is

$$
F_{c,\mathrm{ax}} = \frac{\rho_b A_b\, \omega^2 r_{\mathrm{cm}}^2\, \phi}{2\tan\beta}. \tag{8.27}
$$

Primary Pulley Contribution On the primary pulley, the belt rotates at approximately $\omega = \omega_p$. The outer belt radius equals the primary outer radius from Equation (7.2), which is piecewise in s and accounts for the deadzone behavior.

The mass-centroid radius is

$$r_{p,\mathrm{cm}}(s) = r_{p,\mathrm{outer}}(s) - \Delta r_{\mathrm{cm}}, \tag{8.28}$$

and the wrap angle $\phi_p(s)$ is given by Equation (7.18).

Substituting into Equation (8.27) yields the primary belt centrifugal contribution to axial clamping:

$$F_{c,p}(s, \omega_p) = \frac{\rho_b A_b\, \omega_p^2\, r_{p,\mathrm{cm}}(s)^2\, \phi_p(s)}{2 \tan \beta}. \tag{8.29}$$

Remark (Belt Angular Velocity)

Using the primary angular velocity ω_p in the centrifugal force calculation is an approximation that assumes the belt speed matches the primary sheave speed. In reality, belt slip and compliance may cause the belt to rotate at a vastly different angular velocity than the primary sheave.

Secondary Pulley Contribution On the secondary pulley, the belt rotates at approximately $\omega = \omega_s$. The outer belt radius $r_{s,\mathrm{outer}}(s)$ is determined implicitly by the belt-length constraint (see Equation (7.16)).

The mass-centroid radius is

$$r_{s,\mathrm{cm}}(s) = r_{s,\mathrm{outer}}(s) - \Delta r_{\mathrm{cm}}, \tag{8.30}$$

and the wrap angle $\phi_s(s)$ is given by Equation (7.19).

The secondary belt centrifugal contribution to axial clamping is

$$F_{c,s}(s, \omega_s) = \frac{\rho_b A_b\, \omega_s^2\, r_{s,\mathrm{cm}}(s)^2\, \phi_s(s)}{2 \tan \beta}. \tag{8.31}$$

8.5 Axial Shift Dynamics

Earlier (see Equation (8.1)), the evolution of the shift coordinate was established as

$$m\, \ddot{s} = F_{p,\mathrm{ax}} - F_{s,\mathrm{ax}}, \tag{8.32}$$

where $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ are the axial forces generated by the primary and secondary mechanisms, respectively.

We now complete this model by:

1. Expanding the force terms to include centrifugal belt loading,
2. Replacing the lumped mass m with the equivalent inertia associated with coordinated shift motion.

Equivalent Inertia Associated with s Newton's second law applies to physical masses in translation. Since s parameterizes coordinated motion of several bodies, we construct an equivalent mass m_{eq} such that the total kinetic energy associated with shift can be written in the form of Theory 4 (Kinetic Energy (Translational)):

$$T_{\text{shift}} = \frac{1}{2} m_{\text{eq}} \, \dot{s}^2.$$

The movable primary sheave translates with speed $\dot{s}$ in the positive coordinate direction. The movable secondary sheave translates with equal magnitude speed (but opposite physical direction), so its kinetic energy also contributes proportionally to $\dot{s}^2$.

Let $m_{p,m}$ and $m_{s,m}$ denote the masses of the movable primary and secondary sheaves.

Belt Mass Contribution The total belt mass is

$$m_b = \rho_b A_b L_b, \tag{8.33}$$

where A_b is the trapezoidal cross-sectional area from Equation (8.17) and L_b is the constant belt length.

The belt moves continuously with a single belt speed, but the axial component of that motion associated with a change in s is determined by geometry. From the axial-radial coupling in Equation (7.44), a change in sheave separation s is shared between the two conical faces supporting the belt wedge. Thus, the belt's axial velocity relative to the pulley axis is

$$\dot{s}_b = \frac{\dot{s}}{2}. \tag{8.34}$$

The kinetic energy associated with this axial motion is therefore

$$T_b = \frac{1}{2} m_b \left(\frac{\dot{s}}{2}\right)^2 = \frac{1}{2}\left(\frac{m_b}{4}\right)\dot{s}^2.$$

Expressed in the generalized coordinate s, the belt contributes an effective mass of $m_b/4$.

Total Equivalent Mass The total inertia associated with shift becomes

$$m_{\text{eq}} = m_{p,m} + m_{s,m} + \frac{m_b}{4} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}. \tag{8.35}$$

Net Axial Force in the s Coordinate Collecting the axial force contributions derived previously:

- $F_{p,\text{ax}}(s, \omega_p)$ from Equation (8.10),
- $F_{s,\text{ax}}(s, \tau_s)$ from Equation (8.16),
- $F_{c,p}(s, \omega_p)$ from Equation (8.29),
- $F_{c,s}(s, \omega_s)$ from Equation (8.31).

With the sign convention defined above, the generalized force in the positive s direction is

$$\sum F_s = \Big(F_{p,\mathrm{ax}} + F_{c,p}\Big) - \Big(F_{s,\mathrm{ax}} + F_{c,s}\Big). \tag{8.36}$$

Shift Equation of Motion Applying Newton's second law in the generalized coordinate,

$$\sum F_s = m_{\mathrm{eq}}\,\ddot{s},$$

and substituting Equation (8.35) and Equation (8.36), we obtain

Axial Shift Acceleration **(8.37)**

$$\ddot{s} = \frac{\Big(F_{p,\mathrm{ax}}(s,\omega_p) + F_{c,p}(s,\omega_p)\Big) - \Big(F_{s,\mathrm{ax}}(s,\tau_s) + F_{c,s}(s,\omega_s)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}.$$

Equation (8.37) provides the axial shift acceleration as a function of the current state variables $(s, \dot{s}, \omega_p, \omega_s)$ and the reacted secondary torque τ_s. The remaining coupling between axial and rotational dynamics enters through τ_s, which will be determined later using Theory 2 (Newton's Second Law (Rotation)).

8.6 Section Summary and Forward Outlook

In this section we established a complete axial force model for the CVT mechanism and derived a closed-form equation of motion for the generalized shift coordinate $s(t)$.

At this stage, the axial subsystem is formally closed *except* for one remaining coupling quantity: the reacted secondary torque τ_s. Although τ_s appears explicitly in the secondary axial force model, its value is determined by rotational dynamics of the drivetrain and by torque transmission through the belt. Aside from the shared state variables, the axial and rotational subsystems are coupled only through this torque term.

The next step is to determine how torque is transmitted across the CVT as a function of pulley radii and speed ratio. This will establish the relationship between τ_s, input torque at the primary, and the evolving ratio $R = r_s/r_p$. Once this transmission relationship is obtained, the axial and rotational equations can be combined into a unified, state-space representation of the full CVT dynamics.

With the axial force structure now complete, attention turns to the torque transmission mechanism that closes the remaining dynamical loop.

9 Torque Transmission and Coupling Torque

The axial shift dynamics derived in Section 8.5 require the transferred secondary torque τ_s appearing in Equation (8.16). In this section we determine the torque transmitted through the CVT as a function of the rotational states and the instantaneous ratio $R(t)$.

We proceed in four steps:

1. Write rotational equations of motion for the primary and secondary pulleys.
2. Define the inertias associated with each pulley.
3. Introduce the CVT ratio kinematics.
4. Eliminate accelerations to obtain the transmitted torque.

The result provides the ideal torque transmission law associated with the geometric ratio $R(t)$. This serves as the baseline torque model and will be generalized in the next section.

9.1 System Definition and Torque Path

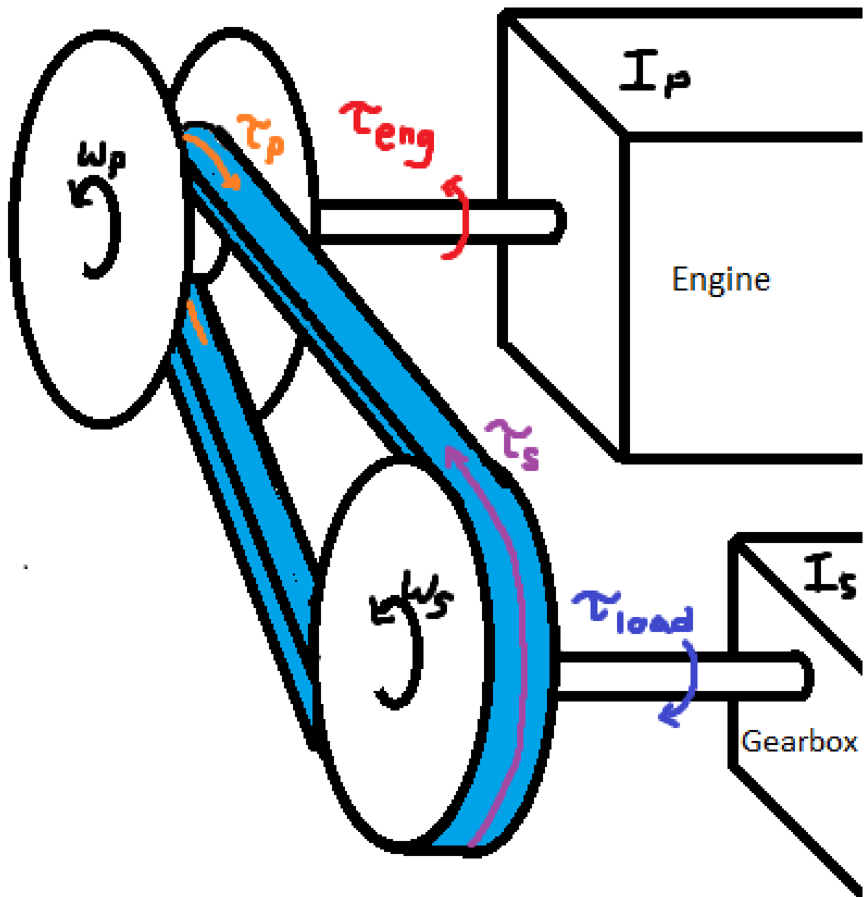

Figure 24: Torque transmission schematic. Engine torque τ_{eng} drives the primary pulley (angular velocity ω_p). Coupling torque τ_p is transmitted through the belt to the secondary pulley (angular velocity ω_s), where it appears as τ_s. The secondary delivers torque to the downstream drivetrain, which resists motion with torque τ_{load}.

We model two rotating bodies:

- Primary pulley (rigidly connected to the engine),
- Secondary pulley (driving the downstream drivetrain and vehicle).

Let

- I_p : total rotational inertia of the engine + primary CVT pulley,
- I_s : total rotational inertia about the secondary pulley,
- $\tau_{\text{eng}}(t)$: engine output torque applied to the primary pulley, modeled as a user-supplied function of primary speed (see Section 9.3),
- $\tau_p(t)$: coupling torque (as seen at the primary pulley),
- $\tau_s(t)$: coupling torque (as seen at the secondary pulley),
- $\tau_{\text{load}}(t)$: resisting torque reflected to the secondary pulley from road loads (see Section 9.4),

9.2 Rotational Equations of Motion

Applying Theory 2 (Newton's Second Law (Rotation)) to each pulley:

Primary Pulley Rotational Dynamics **(9.1)**

$$\tau_{\text{eng}}(t) - \tau_p(t) = I_p\, \alpha_p(t)$$

Secondary Pulley Rotational Dynamics **(9.2)**

$$\tau_s(t) - \tau_{\text{load}}(t) = I_s\, \alpha_s(t)$$

where

$$\alpha_p(t) = \frac{d\omega_p}{dt}, \qquad \alpha_s(t) = \frac{d\omega_s}{dt}.$$

At this stage, the externally applied torques $\tau_{\text{eng}}(t)$ and $\tau_{\text{load}}(t)$ are treated as known inputs to the system. The inertias I_p and I_s are physical constants determined by the drivetrain configuration.

The remaining unknown dynamic quantities are therefore

$$\alpha_p(t), \qquad \alpha_s(t), \qquad \tau_p(t), \qquad \tau_s(t).$$

Equations Equation (9.1) and Equation (9.2) provide two relations among these four unknowns. Two additional relations are therefore required to close the rotational system.

9.3 Engine Torque Model

The engine provides the driving torque $\tau_{\text{eng}}(t)$ acting on the primary pulley.

In this model, engine output torque is represented as a user-supplied function of primary angular velocity. Under the full-throttle assumption (Assumption 15 (Full-Throttle Engine Model)),

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}\big(\omega_p(t)\big), \tag{9.3}$$

so torque depends only on instantaneous primary speed.

This representation is consistent with standard engine characterization, where torque is measured as a function of crankshaft speed under steady full-load conditions.

Admissibility Constraints The supplied torque function must satisfy:

- It is defined over a bounded operating domain $\omega_{\min} \leq \omega_p \leq \omega_{\max}$.
- It is finite and real-valued over this domain.
- Outside this domain, torque is either saturated or ω_p is constrained to remain within bounds.

No specific functional form is required. The dynamical model only requires the mapping $\omega_p \mapsto \tau_{\text{eng}}$ to evaluate Equation (9.1).

Example: Kohler CH440 Engine Curve As an illustrative example, Figure 25 shows a representative torque and power curve for the Kohler CH440 engine.

SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

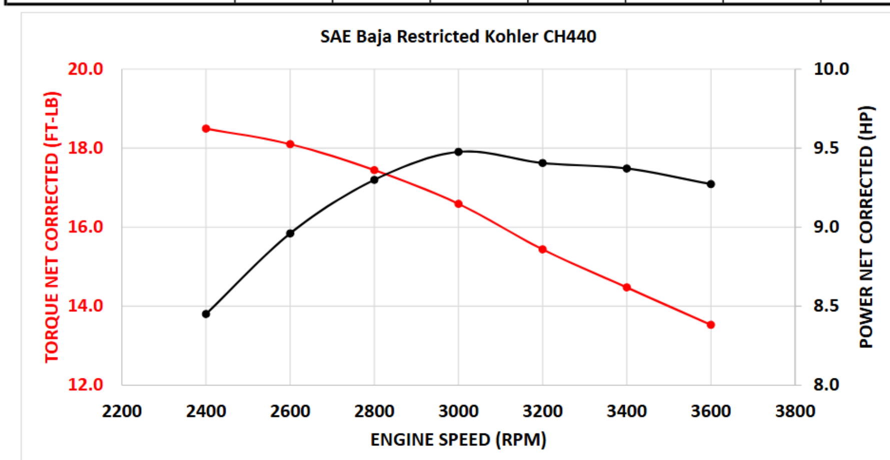

Figure 25: Representative torque and power curves for the Kohler CH440 engine. Torque is supplied as a function of angular velocity and serves as the input mapping in Equation (9.3).

In simulation, the discrete curve is interpolated to provide a continuous admissible torque function over the operating range.

9.4 Load Torque Model

The resisting torque $\tau_{\text{load}}(t)$ in Equation (9.2) represents the longitudinal road load reflected to the secondary pulley.

We proceed in three stages:

1. Express vehicle speed in terms of secondary rotation.
2. Derive the total longitudinal road force.
3. Reflect that force back to the secondary shaft.

Kinematic Relations Let r_w denote effective wheel rolling radius and $G > 0$ the fixed gear ratio defined by

$$\omega_s = G\,\omega_w.$$

Vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G}\,\omega_s(t). \tag{9.4}$$

Longitudinal Road Force Using Theory 1 (Newton's Second Law (Translation)), the total longitudinal resistive force on the vehicle is modeled as

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \tag{9.5}$$

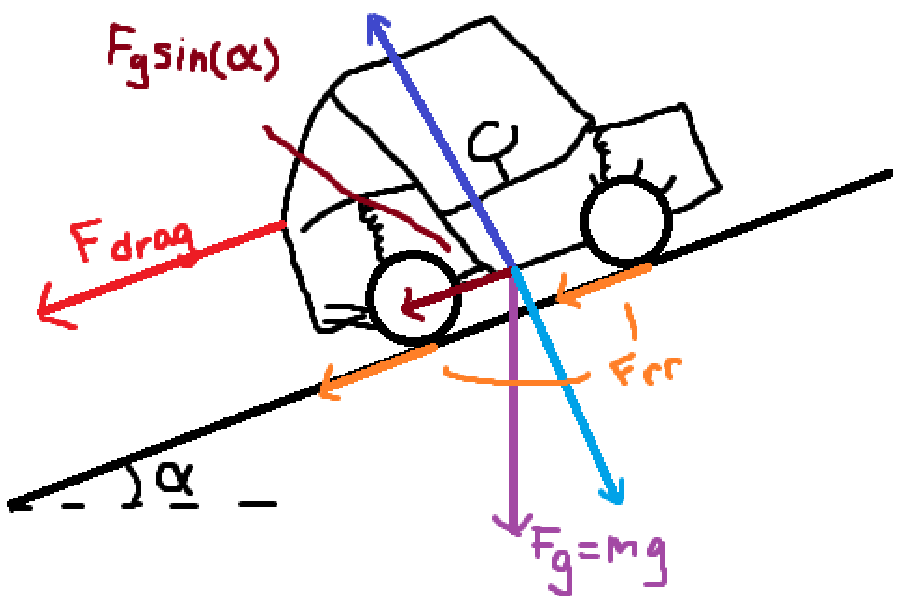

Figure 26: Free-body diagram of a vehicle on an incline. The weight mg resolves into $mg \sin\alpha$ along the slope and $mg\cos\alpha$ normal to the surface. Rolling resistance and aerodynamic drag oppose motion.

Rolling Resistance From Theory 14 (Rolling Resistance),

$$F_{\text{rr}} = C_{\text{rr}}\,N. \tag{9.6}$$

From the incline geometry,

$$N = mg\cos\alpha. \tag{9.7}$$

Thus

$$F_{\mathrm{rr}}(v, \alpha) = C_{\mathrm{rr}} m g \cos\alpha \,\mathrm{sgn}(v). \tag{9.8}$$

Grade Resistance From Theory 15 (Gravitational Force), the gravitational component along the slope is

$$F_{\mathrm{grade}}(\alpha) = m g \sin\alpha. \tag{9.9}$$

Aerodynamic Drag From Theory 13 (Air Resistance (Quadratic Drag)),

$$F_{\mathrm{drag}}(v) = \frac{1}{2}\rho C_d A_f \, v^2 \,\mathrm{sgn}(v). \tag{9.10}$$

Total Road Force Substituting the above components into Equation (9.5) yields

$$\begin{aligned} F_{\mathrm{road}}(t) = {} & C_{\mathrm{rr}} m g \cos\alpha \,\mathrm{sgn}(v(t)) \\ & + m g \sin\alpha \\ & + \frac{1}{2}\rho C_d A_f \, v(t)^2 \,\mathrm{sgn}(v(t))\,. \end{aligned} \tag{9.11}$$

Using Equation (9.4),

$$v(t) = \frac{r_w}{G}\,\omega_s(t),$$

the road force is now fully expressed in terms of the secondary state variable $\omega_s(t)$.

Reflection to Secondary Torque Wheel torque is

$$\tau_{\mathrm{road},w}(t) = r_w F_{\mathrm{road}}(t). \tag{9.12}$$

Assuming ideal power transmission (Assumption 13 (Ideal CVT Power Transmission)), the resisting torque reflected to the secondary pulley is

$$\tau_{\mathrm{load}}(t) = \frac{1}{G}\,\tau_{\mathrm{road},w}(t) = \frac{r_w}{G} F_{\mathrm{road}}(t). \tag{9.13}$$

Substituting Equation (9.11) gives

$$\boxed{\tau_{\mathrm{load}}(t) = \frac{r_w}{G}\left[C_{\mathrm{rr}} m g \cos\alpha \,\mathrm{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big) + m g \sin\alpha + \frac{1}{2}\rho C_d A_f \left(\frac{r_w}{G}\omega_s(t)\right)^2 \mathrm{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big)\right].} \tag{9.14}$$

9.5 Inertia Definitions

The rotational equations Equation (9.1)–Equation (9.2) require the effective rotational inertia about each pulley axis.

Primary Pulley Inertia The primary pulley is rigidly connected to the engine. All components on this side rotate with angular velocity ω_p, so no reflection is required.

The effective inertia is therefore the direct sum

$$I_p = I_{\text{eng}} + I_{\text{prim}}, \tag{9.15}$$

where

- I_{eng} is the engine rotational inertia,
- I_{prim} is the primary CVT pulley inertia.

Secondary Pulley Inertia On the secondary side, all downstream components must be expressed as an equivalent inertia about the secondary axis.

Let

- I_{sec} : secondary CVT pulley inertia,
- I_{gb} : gearbox + intermediate shaft inertia,
- I_{wheel} : wheel + hub inertia about the wheel axis,
- m : total vehicle mass,
- r_w : effective wheel rolling radius,
- G : fixed gear ratio defined by $\omega_s = G\,\omega_w$.

Vehicle Mass as Equivalent Rotational Inertia The vehicle's translational kinetic energy is

$$T_{\text{trans}} = \frac{1}{2}mv^2,$$

with $v = r_w\omega_w$.

Substituting,

$$T_{\text{trans}} = \frac{1}{2}m(r_w\omega_w)^2 = \frac{1}{2}(mr_w^2)\,\omega_w^2.$$

Thus the vehicle mass behaves as an effective rotational inertia

$$I_{\text{veh,wheel}} = mr_w^2$$

about the wheel axis.

Reflection to the Secondary Axis Because $\omega_s = G\,\omega_w$, we have $\omega_w = \omega_s/G$.

Expressing the downstream kinetic energy in terms of ω_s:

$$T = \frac{1}{2} I_{\text{wheel}} \left(\frac{\omega_s}{G}\right)^2 + \frac{1}{2}(m r_w^2) \left(\frac{\omega_s}{G}\right)^2.$$

Factoring,

$$T = \frac{1}{2} \left(\frac{I_{\text{wheel}} + m r_w^2}{G^2}\right) \omega_s^2.$$

Therefore, inertias at the wheel reflect to the secondary pulley as

$$I_{\text{reflected}} = \frac{I_{\text{wheel}} + m r_w^2}{G^2}.$$

Total Secondary Inertia The total effective inertia about the secondary pulley is

$$I_s = I_{\text{sec}} + I_{\text{gb}} + \frac{I_{\text{wheel}} + m r_w^2}{G^2}. \tag{9.16}$$

Equation Equation (9.16) defines the total inertia opposing angular acceleration of the secondary pulley.

9.6 Ratio Kinematics

The CVT ratio was defined geometrically in Section 7.9 (see Equation (7.22)) as

$$R(t) = \frac{r_{s,\text{eff}}(t)}{r_{p,\text{eff}}(t)}.$$

To relate pulley speeds, we now introduce the *no-slip constraint.*

Under ideal belt transmission, the belt surface speed must match the tangential speed of both pulley faces. This enforces

$$r_{p,\text{eff}}(t)\,\omega_p(t) = r_{s,\text{eff}}(t)\,\omega_s(t). \tag{9.17}$$

Substituting the geometric ratio definition gives

$$\omega_p(t) = R(t)\,\omega_s(t). \tag{9.18}$$

This relation is equivalent to assuming zero slip between belt and sheaves. Slip effects will be handled in a later section.

Differentiating Equation (9.18) with respect to time gives

$$\alpha_p(t) = R(t)\,\alpha_s(t) + \omega_s(t)\,\dot{R}(t), \tag{9.19}$$

where $\dot{R}(t) = \dfrac{dR}{dt}$ (see Equation (7.50)).

Equation Equation (9.19) captures the dynamic coupling introduced by a time-varying ratio: even if $\alpha_s = 0$, a changing ratio ($\dot{R} \neq 0$) produces primary angular acceleration.

This provides one of the additional relations required to close the rotational system.

9.7 Torque Mapping Across the CVT

The belt transfers mechanical power between the primary and secondary pulleys. In general,

$$\tau_p(t)\,\omega_p(t) = \tau_s(t)\,\omega_s(t) + P_{\text{loss}}(t), \tag{9.20}$$

where P_{loss} represents internal dissipation (e.g., belt hysteresis, bearing drag).

Following Assumption 13 (Ideal CVT Power Transmission), we neglect these non-slip-related losses and set

$$P_{\text{loss}}(t) = 0.$$

This assumption removes internal inefficiencies but does *not* by itself impose the no-slip constraint.

When the belt is in sticking contact, the kinematic relation Equation (9.18) holds. Substituting $\omega_p = R\,\omega_s$ into the ideal power balance gives

$$\tau_s(t) = R(t)\,\tau_p(t). \tag{9.21}$$

Equation Equation (9.21) is therefore the ideal torque mapping associated with the geometric ratio. If slip occurs, Equation (9.18) no longer holds and the torque relation must be modified accordingly.

9.8 Eliminating Accelerations

We now have four equations (Equation (9.1), Equation (9.2), Equation (9.19), Equation (9.21)) and four unknowns (α_p, α_s, τ_p, τ_s). The goal is to eliminate the accelerations α_p and α_s to obtain a closed-form expression for the coupling torque τ_p in terms of the known quantities τ_{eng}, τ_{load}, I_p, I_s, R, and $\dot{R}$.

From Equation (9.1),

$$\tau_p = \tau_{\text{eng}} - I_p\,\alpha_p.$$

Substituting Equation (9.19):

$$\tau_p = \tau_{\text{eng}} - I_p\big(R\alpha_s + \omega_s\dot{R}\big). \tag{9.22}$$

From Equation (9.2) and Equation (9.21),

$$\alpha_s = \frac{R\tau_p - \tau_{\text{load}}}{I_s}.$$

Substituting into Equation (9.22):

$$\begin{aligned}\tau_p &= \tau_{\text{eng}} - I_p \left[R \left(\frac{R\tau_p - \tau_{\text{load}}}{I_s} \right) + \omega_s \dot{R} \right] \\ &= \tau_{\text{eng}} - \frac{I_p}{I_s} \left(R^2 \tau_p - R\tau_{\text{load}} \right) - I_p \omega_s \dot{R}. \end{aligned} \tag{9.23}$$

Collecting terms:

$$\tau_p \left(1 + \frac{I_p}{I_s} R^2 \right) = \tau_{\text{eng}} + \frac{I_p}{I_s} R\tau_{\text{load}} - I_p \omega_s \dot{R}. \tag{9.24}$$

9.9 Closed-Form Coupling Torque

Solving Equation (9.24):

Coupling Torque (Primary Side) **(9.25)**

$$\tau_p(t) = \frac{\tau_{\text{eng}}(t) + \dfrac{I_p}{I_s} R(t)\, \tau_{\text{load}}(t) - I_p\, \omega_s(t)\, \dot{R}(t)}{1 + \dfrac{I_p}{I_s} R(t)^2}.$$

The coupling torque as seen at the secondary pulley, required by Equation (8.16), then follows from Equation (9.21):

Coupling Torque (Secondary Side) **(9.26)**

$$\tau_s(t) = R(t)\, \tau_p(t).$$

9.10 Section Summary

We began with the rotational equations of motion for the primary and secondary pulleys, introducing four unknowns (α_p, α_s, τ_p, τ_s) but only two dynamical equations.

Closure was achieved by imposing two additional physical relations:

- The *no-slip kinematic constraint* $\omega_p = R\,\omega_s$, together with its time derivative $\alpha_p = R\alpha_s + \omega_s \dot{R}$.
- The *ideal power-transfer assumption* (Assumption 13 (Ideal CVT Power Transmission)), which neglects non-slip-related losses and yields the torque mapping $\tau_s = R\,\tau_p$.

Together, these relations close the rotational system and produce the explicit coupling torque Equation (9.25), which expresses the transmitted torque in terms of the instantaneous ratio $R(t)$ and ratio rate $\dot{R}(t)$.

Substitution of $\tau_s = R\,\tau_p$ into the secondary axial force model Equation (8.16) completes the dynamical coupling required for the axial shift equation Equation (8.37).

10 Traction Limit and Slip Condition

The torque transmission law derived previously assumes ideal no-slip belt traction. Under that assumption, the coupling torque Equation (9.25) (and equivalently Equation (9.26)) represents the torque actually transmitted through the CVT.

In reality, belt-pulley friction is finite. For a given normal clamping force, there exists a maximum tangential traction force that the contact can sustain. If the demanded coupling torque exceeds this traction capacity, slip occurs and the transmitted torque saturates at a friction-limited value.

The objective of this section is to derive a traction inequality of the form

$$|\tau| \leq \tau_{\max}, \tag{10.1}$$

which will later be enforced in place of the ideal coupling law whenever the no-slip solution violates this bound.

10.1 Tight–Slack Tension Difference and Torque Capacity

Consider the wrapped pulley shown in Figure 27. If torque is transmitted from the pulley to the belt, the belt tension cannot be the same on both spans. One side must carry a higher tension than the other. We therefore define

$$\Delta T \;\equiv\; T_{\text{tight}} - T_{\text{slack}}, \tag{10.2}$$

where T_{tight} and T_{slack} are the belt tensions on the two straight spans shown in the figure.

This tension difference arises because the pulley applies tangential traction to the belt along the wrap angle ϕ. The cumulative effect of this traction increases the belt tension progressively from the slack side to the tight side.

At effective contact radius r_{eff}, the transmitted torque is

$$\tau = r_{\text{eff}}\,\Delta T, \qquad |\tau| = r_{\text{eff}}\,|\Delta T|. \tag{10.3}$$

Thus, limiting the transmitted torque is equivalent to limiting the admissible tension difference ΔT.

Under Assumption 4 (Idealized Belt–Pulley Friction Law) and Assumption 5 (Constant Effective Friction Coefficient), the belt–pulley interface obeys a Coulomb friction law with constant coefficient μ. The tangential traction at any point along the wrap is therefore bounded by the local normal load:

$$|F_f| \leq \mu N.$$

The *maximum* admissible tension difference occurs when friction is fully mobilized everywhere along the wrap — that is, at impending slip. Beyond this point, additional tension rise cannot be supported and sliding must occur.

We therefore define a maximum supported tension difference ΔT_{max}, corresponding to the friction-saturated state, and the associated torque capacity

$$\tau_{\text{max}} = r_{\text{eff}}\, \Delta T_{\text{max}}. \tag{10.4}$$

The next subsection computes ΔT_{max} by examining the incremental change in belt tension along the wrap.

Remark (Friction)

TODO: Discuss other options for modelling friction of the rubber belt, reference that paper. Appendix?

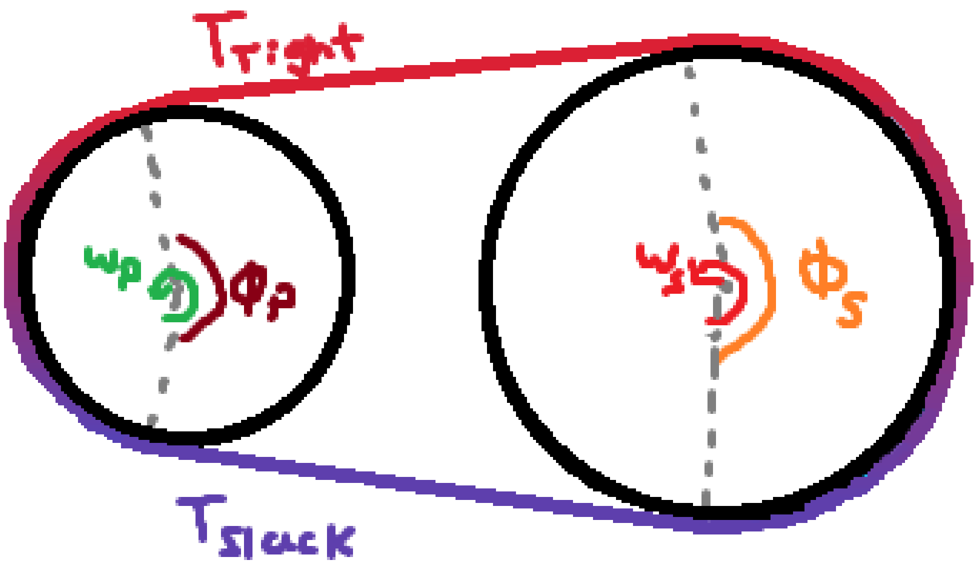

Figure 27: Torque transmission requires a tight–slack tension difference across the wrap.

10.2 Differential Belt Element and Tangential Force Balance

To determine the maximum admissible tension difference ΔT_{max} introduced in Section 10.1, we now examine how belt tension changes *locally* as the belt passes around the pulley.

Specifically, we seek an expression for the incremental tension change $\mathrm{d}T$ over a differential wrap angle $\mathrm{d}\theta$. Once $\mathrm{d}T$ is related to the available friction at the interface, integration over the total wrap angle ϕ will yield ΔT_{max}.

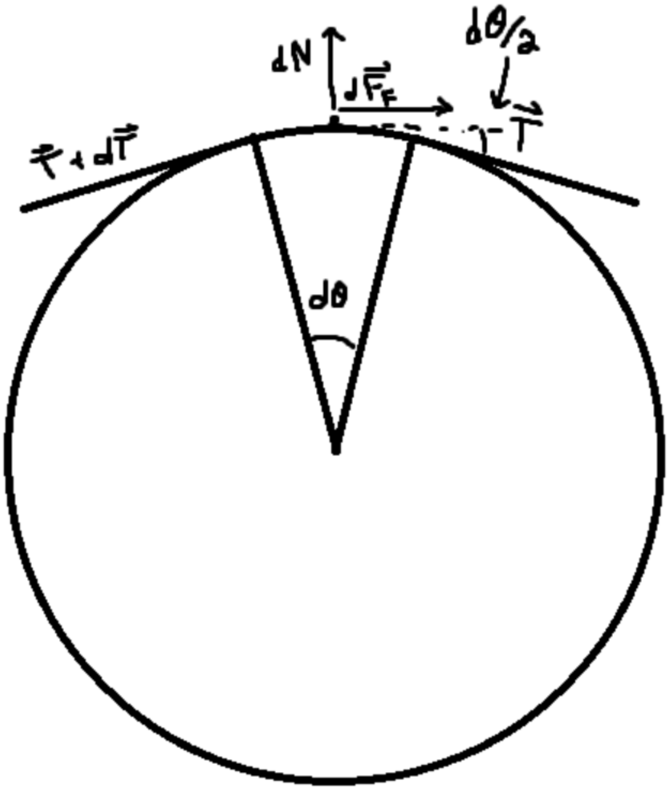

Figure 28: Free-body diagram of a differential belt element spanning $\mathrm{d}\theta$.

Consider the differential belt element shown in Figure 28, spanning an infinitesimal wrap angle $\mathrm{d}\theta$. The belt tension entering the element is T, and leaving is $T + \mathrm{d}T$.

The pulley applies to this element:

- a differential normal force $\mathrm{d}N$ (radial),
- a differential tangential friction force $\mathrm{d}F_f$.

The positive tangential direction is taken along the midpoint tangent of the element, consistent with increasing wrap angle.

Tangential Newton's Second Law Applying Theory 1 (Newton's Second Law (Translation)) in the tangential direction:

$$\sum F_t = \mathrm{d}m\, a_t, \tag{10.5}$$

where a_t is the tangential acceleration of the belt material at the contact point.

The tangential components of the end tensions follow directly from geometry:

$$(T)_t = T\cos\left(\frac{\mathrm{d}\theta}{2}\right), \tag{10.6}$$

$$(T + \mathrm{d}T)_t = (T + \mathrm{d}T)\cos\left(\frac{\mathrm{d}\theta}{2}\right). \tag{10.7}$$

The left-end tension acts in the positive tangential direction, while the right-end tension acts in the negative tangential direction. Including friction, the tangential force balance becomes

$$T\cos\left(\frac{\mathrm{d}\theta}{2}\right) - (T + \mathrm{d}T)\cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_f = \mathrm{d}m\, a_t. \tag{10.8}$$

Simplifying the first two terms exactly,

$$-\,\mathrm{d}T\cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_f = \mathrm{d}m\, a_t. \tag{10.9}$$

Rearranging,

$$\mathrm{d}T\cos\left(\frac{\mathrm{d}\theta}{2}\right) = \mathrm{d}F_f - \mathrm{d}m\, a_t. \tag{10.10}$$

Belt mass element From Equation (8.24),

$$\mathrm{d}m = \rho_b A_b r_{\mathrm{cm}}\, \mathrm{d}\theta. \tag{10.11}$$

so

$$\mathrm{d}m\, a_t = \rho_b A_b r_{\mathrm{cm}} a_t\, \mathrm{d}\theta. \tag{10.12}$$

Substituting into Equation (10.10),

$$\mathrm{d}T\cos\left(\frac{\mathrm{d}\theta}{2}\right) = \mathrm{d}F_f - \rho_b A_b r_{\mathrm{cm}} a_t\, \mathrm{d}\theta. \tag{10.13}$$

Friction limit and impending slip As established in Section 10.1, the maximum admissible tension rise occurs when friction is fully mobilized along the wrap (impending slip). At the differential level, this corresponds to

$$\mathrm{d}F_f = \mu\, \mathrm{d}N. \tag{10.14}$$

Substituting Equation (10.14) into Equation (10.13) yields the maximum local tension increase supported without sliding:

$$\mathrm{d}T_{\max} = \frac{\mu\, \mathrm{d}N - \rho_b A_b r_{\mathrm{cm}} a_t\, \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{10.15}$$

10.3 Integration Across the Wrap

The tight–slack tension difference introduced in Section 10.1 is obtained by accumulating the incremental tension changes $\mathrm{d}T$ along the contact arc. Since tension evolves continuously around the wrap,

$$\Delta T = T_{\text{tight}} - T_{\text{slack}} = \int_{-\phi/2}^{\phi/2} \mathrm{d}T. \tag{10.16}$$

Accordingly, the maximum admissible tension difference is obtained by integrating the local maximum increment $\mathrm{d}T_{\max}$ from Equation (10.15):

$$\Delta T_{\max} = \int_{-\phi/2}^{\phi/2} \frac{\mu\ \mathrm{d}N - \rho_b A_b r_{\text{cm}} a_t\ \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{10.17}$$

In the differential limit $\mathrm{d}\theta \to 0$, $\cos(\mathrm{d}\theta/2) \to 1$, yielding

$$\Delta T_{\max} = \int_{-\phi/2}^{\phi/2} \mu\ \mathrm{d}N - \int_{-\phi/2}^{\phi/2} \rho_b A_b r_{\text{cm}} a_t\ \mathrm{d}\theta. \tag{10.18}$$

The first integral represents the total normal load supported over the wrap, which we denote

$$N_\phi \ \equiv \ \int_{-\phi/2}^{\phi/2} \mathrm{d}N. \tag{10.19}$$

(The explicit relationship between N_ϕ and the axial clamping force is derived in Section 10.5.)

From Assumption 9 (Uniform Contact Radius Around Wrap), r_{cm} and a_t are uniform over the wrap, and so the second integral simplifies to $\rho_b A_b r_{\text{cm}} a_t \phi$, giving

$$\Delta T_{\max} = \mu N_\phi - \rho_b A_b r_{\text{cm}} a_t \phi. \tag{10.20}$$

This leaves two unknowns: the normal load N_ϕ and the tangential acceleration a_t.

10.4 Tangential Acceleration of a Belt Element on a Circular Wrap

This subsection derives the tangential acceleration term required in the local traction inequality in Equation (10.20). In that inequality we apply Newton's second law *along the direction of belt motion*, so we need an explicit expression for the belt element's tangential acceleration a_t.

Goal For a belt element of differential mass $\mathrm{d}m$ moving along the wrap, the tangential force balance uses

$$\sum F_t = \mathrm{d}m\, a_t,$$

so our goal is to compute a_t from the belt kinematics when both *angular speed* and *wrap radius* may vary:

$$\omega(t) = \dot{\theta}(t), \qquad \alpha(t) = \dot{\omega}(t), \qquad r(t), \qquad \dot{r}(t).$$

Coordinate system and unit vectors Because we care about forces *along the belt*, we choose a basis aligned with the belt's local geometry:

- $\mathbf{e}_r(t)$ points radially outward from the pulley center O to the belt point P,
- $\mathbf{e}_\theta(t)$ is the unit tangent, obtained by rotating $\mathbf{e}_r$ by $+90°$ in the direction of increasing θ.

These are not arbitrary symbols: they are *the* directions of radial and tangential motion on a circle, and the belt's local tangential direction is exactly $\mathbf{e}_\theta$.

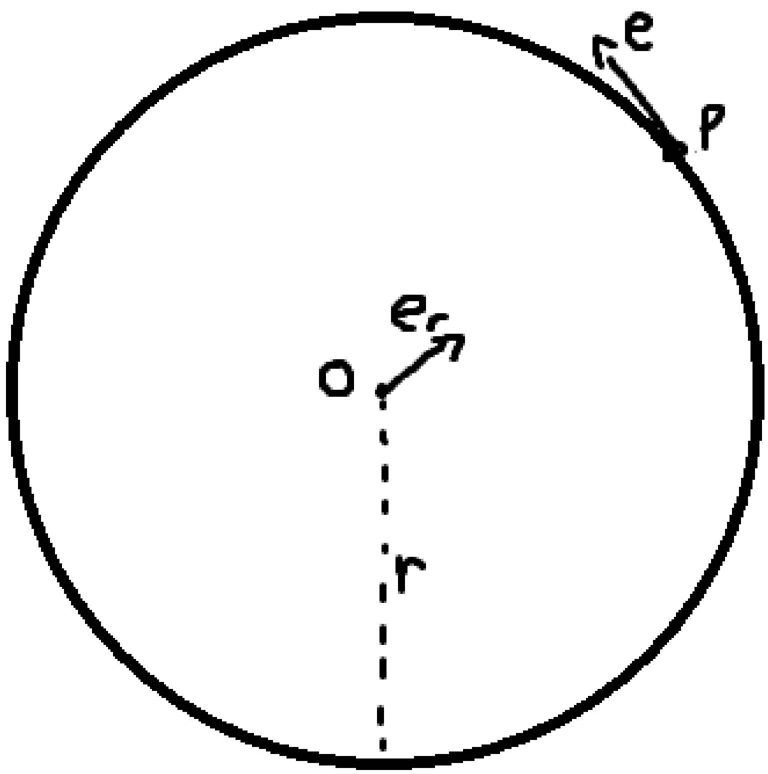

Figure 29: Definition of the radial and tangential unit vectors.

Work backwards: to get a_t, compute $\mathbf{a}$ and take its tangential component Tangential acceleration is the component of the acceleration vector along $\mathbf{e}_\theta$:

$$a_t \equiv \mathbf{a} \cdot \mathbf{e}_\theta. \tag{10.21}$$

So we will compute $\mathbf{a} = \dot{\mathbf{v}}$, then read off the $\mathbf{e}_\theta$ component.

Step 1: position and velocity The position of P from the center is

$$\mathbf{r}(t) = r(t)\,\mathbf{e}_r(t). \tag{10.22}$$

Differentiate Equation (10.22) using the product rule:

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\,\mathbf{e}_r + r\,\dot{\mathbf{e}}_r. \tag{10.23}$$

This shows why we must track how the unit vectors change with time: even if r were constant, $\mathbf{e}_r$ rotates as the belt moves around the pulley.

Step 2: derive $\dot{\mathbf{e}}_r$ and $\dot{\mathbf{e}}_\theta$ (no assumptions) Over a small change $\mathrm{d}\theta$, the unit vectors rotate rigidly by the same angle. The derivative with respect to angle is therefore:

$$\frac{\mathrm{d}\mathbf{e}_r}{\mathrm{d}\theta} = \mathbf{e}_\theta, \qquad \frac{\mathrm{d}\mathbf{e}_\theta}{\mathrm{d}\theta} = -\mathbf{e}_r. \tag{10.24}$$

(Geometric justification: increasing θ rotates $\mathbf{e}_r$ toward the tangent; increasing θ rotates the tangent toward $-\mathbf{e}_r$.)

Convert angle-derivatives to time-derivatives with the chain rule $\mathrm{d}(\cdot)/\,\mathrm{d}t = (\,\mathrm{d}(\cdot)/\,\mathrm{d}\theta)\,\dot{\theta}$ and $\omega = \dot{\theta}$:

$$\dot{\mathbf{e}}_r = \omega\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\omega\,\mathbf{e}_r. \tag{10.25}$$

Step 3: velocity in radial–tangential form Substitute $\dot{\mathbf{e}}_r$ from Equation (10.25) into Equation (10.23):

$$\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\omega\,\mathbf{e}_\theta. \tag{10.26}$$

This is already intuitive: $\dot{r}$ is the radial speed, and $r\omega$ is the tangential speed.

Step 4: acceleration and its tangential component Differentiate Equation (10.26):

$$\begin{aligned}\mathbf{a} = \dot{\mathbf{v}} &= \frac{\mathrm{d}}{\mathrm{d}t}(\dot{r}\,\mathbf{e}_r) + \frac{\mathrm{d}}{\mathrm{d}t}(r\omega\,\mathbf{e}_\theta)\\ &= \ddot{r}\,\mathbf{e}_r + \dot{r}\,\dot{\mathbf{e}}_r + (\dot{r}\,\omega + r\alpha)\mathbf{e}_\theta + r\omega\,\dot{\mathbf{e}}_\theta.\end{aligned} \tag{10.27}$$

Now substitute the unit-vector time derivatives from Equation (10.25):

$$\dot{\mathbf{e}}_r = \omega\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\omega\,\mathbf{e}_r,$$

and collect like directions:

$$\begin{aligned}\mathbf{a} &= \ddot{r}\,\mathbf{e}_r + \dot{r}(\omega\,\mathbf{e}_\theta) + (\dot{r}\,\omega + r\alpha)\mathbf{e}_\theta + r\omega(-\omega\,\mathbf{e}_r)\\ &= \big(\ddot{r} - r\omega^2\big)\mathbf{e}_r + \big(r\alpha + 2\dot{r}\,\omega\big)\mathbf{e}_\theta.\end{aligned} \tag{10.28}$$

Finally, by the definition Equation (10.21), the tangential acceleration is the $\mathbf{e}_\theta$ component:

$$a_t = r\alpha + 2\dot{r}\,\omega. \tag{10.29}$$

Where the factor of 2 comes from The two $\dot{r}\,\omega$ contributions in Equation (10.28) come from *two distinct effects*:

- $\dot{r}\,\dot{\mathbf{e}}_r$: even the *radial* velocity $\dot{r}\,\mathbf{e}_r$ gains a tangential contribution because $\mathbf{e}_r$ rotates at rate ω;
- $\frac{\mathrm{d}}{\mathrm{d}t}(r\omega)$: the tangential speed $r\omega$ changes if r changes.

Adding these yields the total $2\dot{r}\,\omega$ term.

10.5 Normal Force from Axial Clamping

The traction bound Equation (10.20) depends on the total normal load over the wrap,

$$N_\phi \equiv \int_{-\phi/2}^{\phi/2} \mathrm{d}N.$$

We now express N_ϕ in terms of the axial clamping force generated by the pulley mechanism.

Axial clamp produces radial normal load The belt is pressed between two opposing conical sheaves. An axial clamping force F_{ax} pushes the sheaves together. Because the sheave faces are inclined at cone half-angle β, this axial force generates a radial normal load on the belt.

From the conical kinematics established in Equation (7.1),

$$\frac{\mathrm{d}r}{\mathrm{d}s} = \frac{1}{2\tan\beta}.$$

To relate forces consistently, we use virtual work. The incremental work done by the axial clamp must equal the incremental work associated with radial compression:

$$F_{\mathrm{ax}}\,\mathrm{d}s = F_{\mathrm{rad}}\,\mathrm{d}r.$$

Substituting $\mathrm{d}r = (\mathrm{d}r/\mathrm{d}s)\,\mathrm{d}s$ gives

$$F_{\mathrm{rad}} = \frac{F_{\mathrm{ax}}}{\mathrm{d}r/\mathrm{d}s} = 2F_{\mathrm{ax}}\tan\beta. \tag{10.30}$$

Thus, the axial clamp force produces a total radial normal load

$$F_{\mathrm{rad}} = 2F_{\mathrm{ax}}\tan\beta. \tag{10.31}$$

Identification with the wrap-normal load The radial load F_{rad} derived above represents the total inward force applied by the two sheave faces onto the belt. That load is carried entirely by the distributed belt–pulley contact forces $\mathrm{d}N$ along the wrap.

Since no other radial forces act on the belt within the contact region, radial equilibrium requires that the sum of the distributed contact forces equal the total clamp-induced radial load. Therefore,

$$\boxed{N_\phi = F_{\mathrm{rad}} = 2F_{\mathrm{ax}}\tan\beta.} \tag{10.32}$$

This result states that the total normal load available for traction is set directly by the axial clamping force through the cone geometry.

10.6 Torque Capacity Inequality

From Section 10.1, the maximum transmissible torque is determined by the maximum admissible tension difference:

$$\tau_{\max} = r_{\text{eff}}\, \Delta T_{\max}. \tag{10.33}$$

Using the integrated result from Equation (10.20),

$$\Delta T_{\max} = \mu N_\phi - \rho_b A_b r_{\text{cm}} a_t \phi,$$

the traction limit becomes

$$|\tau| \le r_{\text{eff}} \left(\mu N_\phi - \rho_b A_b r_{\text{cm}} a_t \phi\right). \tag{10.34}$$

Substituting the clamp–normal relation from Equation (10.32),

$$N_\phi = 2F_{\text{ax}} \tan\beta,$$

gives

$$|\tau| \le r_{\text{eff}} \left(\mu(2F_{\text{ax}} \tan\beta) - \rho_b A_b r_{\text{cm}} a_t \phi\right). \tag{10.35}$$

Finally, substituting the belt tangential acceleration from Equation (10.29),

$$a_t = r\alpha + 2\dot{r}\,\omega,$$

yields the explicit traction-limited torque inequality

Traction-Limited Torque Capacity **(10.36)**

$$|\tau| \le r_{\text{eff}} \left[2\mu F_{\text{ax}} \tan\beta - \rho_b A_b r_{\text{cm}} \phi \left(r\alpha + 2\dot{r}\,\omega\right)\right].$$

Remark (Why this differs from the classical capstan exponential)

In the classical capstan problem, the normal load is generated *by the belt tension* itself (a larger tension implies a proportionally larger contact normal), which leads to the exponential relation $T_1/T_2 = e^{\mu\phi}$.

Here, the normal load is imposed externally by the clamping mechanism: N_ϕ is determined by F_{ax} through Equation (10.32). Therefore, the available traction scales *linearly* with clamping force, not exponentially with belt tension. Centrifugal loading and ratio change enter additionally through the inertial term in Equation (10.36).

Remark (Torque-Reactive Clamping and Implicit Capacity at the Secondary)

For the secondary pulley, the specific axial clamp force F_{ax} used in this model depends on the transmitted torque τ_s (see Equation (8.16)). However, the traction capacity $\tau_{\max}$ derived above depends on F_{ax} through Equation (10.32) and Equation (10.36).
Consequently, at the secondary interface the admissible torque satisfies an implicit relation of the form

$$|\tau_s| \le \tau_{\max}\big(F_{\text{ax}}(\tau_s)\big).$$

In other words, the transmitted torque influences the clamp force, the clamp force determines the traction capacity, and that capacity bounds the transmitted torque. This circular dependence arises only because the secondary mechanism is torque-reactive.
For the specific secondary model adopted in this work, this implicit inequality can be reduced to a closed-form solution (see Appendix D). The explicit algebra is omitted here to preserve clarity, as it depends on the detailed functional form of $F_{\text{ax}}(s, \tau_s)$ and is not required to understand the main traction derivation.

10.7 Section Conclusion and Transition

This section derived a traction-limited torque bound for a single pulley.

Equation Equation (10.36) gives the maximum torque that can be transmitted at the current operating state. It depends only on:

- axial clamping force F_{ax},
- cone angle β,
- wrap angle ϕ,
- belt mass and kinematics.

No exponential amplification occurs because the normal force is externally imposed and bounded.

The ideal coupling torque derived earlier remains valid only when it satisfies Equation (10.36). If the demanded torque exceeds this bound, the belt cannot sustain static friction and slip must occur.

The next section defines the stick and slip regimes and modifies the governing equations accordingly.

11 Slip Regimes and Traction-Limited Coupling Torque

Earlier sections produced two results:

- A no-slip (rigid-traction) coupling torque, denoted τ_{ns}, which is the torque required to enforce perfect belt adherence.
- A traction capacity bound of the form $|\tau| \le \tau_{\max}$, derived in Section 10.6.

The purpose of this section is to combine these into a single physically admissible coupling torque law that:

1. reduces to the no-slip model when feasible,
2. never violates the traction bound,
3. and produces sliding torque that opposes relative motion when slip occurs.

11.1 Slip Metric: Detecting Stick and Slip

Slip is defined at the belt–pulley contact. If the belt adheres to both pulleys, the tangential surface speeds must match.

Define the contact speeds

$$v_p = r_{p,\text{eff}}\,\omega_p, \qquad v_s = r_{s,\text{eff}}\,\omega_s, \tag{11.1}$$

and the relative mismatch

$$v_{\text{rel}} = v_p - v_s = r_{p,\text{eff}}\,\omega_p - r_{s,\text{eff}}\,\omega_s. \tag{11.2}$$

Stick condition:

$$v_{\text{rel}} = 0. \tag{11.3}$$

Slip condition:

$$v_{\text{rel}} \neq 0. \tag{11.4}$$

Thus, v_{rel} is the diagnostic quantity: perfect adherence requires zero relative surface speed.

11.2 Feasibility of the No-Slip Torque

The no-slip model specifies a required coupling torque

$$\tau_{\text{ns}}, \tag{11.5}$$

which is the torque needed to enforce $v_{\text{rel}} = 0$ under the current state.

However, from Section 10.6, the belt–pulley interface cannot transmit a torque exceeding $\tau_{\max}$ in magnitude. Therefore, static adherence is only physically possible if

$$|\tau_{\text{ns}}| \leq \tau_{\max}. \tag{11.6}$$

If Equation (11.6) holds, static friction can enforce $v_{\text{rel}} = 0$ and the system remains in stick.

If instead

$$|\tau_{\text{ns}}| > \tau_{\max}, \tag{11.7}$$

the required torque exceeds the available traction. Static friction cannot maintain adherence, and slip must occur.

11.3 Traction-Limited Coupling Torque Law

The coupling torque must satisfy two physical facts:

1. Static friction can enforce $v_{\text{rel}} = 0$ only if $|\tau_{\text{ns}}| \leq \tau_{\max}$.
2. Once sliding occurs ($v_{\text{rel}} \neq 0$), the interface is in kinetic friction and the transmitted torque remains saturated at $\tau_{\max}$ until the relative speed vanishes.

This leads to the following regime definition.

Stick (static friction). If

$$v_{\text{rel}} = 0 \quad \text{and} \quad |\tau_{\text{ns}}| \le \tau_{\max}, \tag{11.8}$$

then static traction can be maintained and

$$\tau_c = \tau_{\text{ns}}. \tag{11.9}$$

Slip (kinetic friction). If

$$v_{\text{rel}} \neq 0 \quad \text{or} \quad |\tau_{\text{ns}}| > \tau_{\max}, \tag{11.10}$$

then the transmitted torque is friction-limited and opposes the relative motion:

$$\tau_c = \tau_{\max}\,\text{sign}(v_{\text{rel}}). \tag{11.11}$$

Physical interpretation. While $v_{\text{rel}} \neq 0$, the interface is in sliding contact. The friction torque remains saturated at $\tau_{\max}$ and always acts to oppose the mismatch.

If $|\tau_{\text{ns}}| < \tau_{\max}$ during slip, the saturated friction torque reduces $|v_{\text{rel}}|$ until it reaches zero, at which point stick may be re-established.

If instead $|\tau_{\text{ns}}| > \tau_{\max}$, the torque imbalance continues to increase the mismatch, and slip persists.

The complete traction-limited coupling law is therefore

Traction-Limited Coupling Torque **(11.12)**

$$\boxed{\tau_c = \begin{cases} \tau_{\text{ns}}, & v_{\text{rel}} = 0 \text{ and } |\tau_{\text{ns}}| \le \tau_{\max}, \\ \tau_{\max}\,\text{sign}(v_{\text{rel}}), & \text{otherwise.} \end{cases}}$$

Remark (Numerical regularization)

The discontinuous $\text{sign}(\cdot)$ function in Equation (11.12) may be replaced by a smooth approximation for numerical integration, such as

$$\tau_c = \tau_{\max} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right),$$

where $v_{\text{smooth}} > 0$ controls transition sharpness. This modification improves numerical stability without significantly altering the underlying physical model.

12 Complete CVT Dynamic Model

This section assembles the results of the preceding derivations into a single closed dynamical system. No new physics is introduced. The purpose is to state (i) the chosen state variables, (ii) the intermediate quantities computed from those states (geometry, ratio, traction limits), and (iii) the governing ODEs.

12.1 State Definition

The dynamic state vector is

$$\mathbf{x} = \big(\omega_p,\ \omega_s,\ s,\ \dot{s}\big), \tag{12.1}$$

where ω_p and ω_s are the primary and secondary pulley angular velocities, and s is the axial shift coordinate.

External inputs are the engine torque model $\tau_{\text{eng}}(t)$ and the road-load torque model $\tau_{\text{load}}(t)$.

12.2 Geometric Closure (Ratio and Ratio Rate)

All pulley radii, wrap angles, and belt geometry are determined by s via the belt-length constraint and conical kinematics (see Section 7.11). In particular, the CVT ratio is Equation (7.22):

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{12.2}$$

Its time derivative is given by Equation (7.50) and depends on $(s, \dot{s})$:

$$\dot{R} = \dot{R}(s, \dot{s}). \tag{12.3}$$

These are the only places where $\dot{s}$ enters the ratio kinematics explicitly.

12.3 Coupling Torques and What the Model Solves For

We retain the two-sided torque notation used throughout:

- $\tau_p(t)$: coupling torque as seen at the *primary* pulley,
- $\tau_s(t)$: coupling torque as seen at the *secondary* pulley.

These are not prescribed inputs. They are determined by:

1. a *no-slip* torque demand (from the rotational coupling derivation),
2. a *traction capacity* bound (from the friction derivation),
3. a stick–slip rule that selects the admissible transmitted torque.

When no slip is feasible, the no-slip torque demand is Equation (9.25):

$$\tau_{\text{ns}}(t) \ \triangleq\ \tau_p(t) = \frac{\tau_{\text{eng}}(t) + \dfrac{I_p}{I_s} R(t)\, \tau_{\text{load}}(t) - I_p\, \omega_s(t)\, \dot{R}(t)}{1 + \dfrac{I_p}{I_s} R(t)^2}. \tag{12.4}$$

Torque on the secondary side is related by the ideal geometric mapping (derived from power balance under the ideal transmission assumption):

$$\tau_s(t) = R(t)\, \tau_p(t). \tag{12.5}$$

In slip, the actual transmitted torque is limited by the traction capacity. We denote the instantaneous admissible magnitude by $\tau_{\max}$, computed from Equation (10.36) evaluated on each pulley and combined by the chosen limiting rule (e.g., min across interfaces).

Finally, the *actual* coupling torque used in the dynamics is selected by the traction-limited law Equation (11.12):

$$\tau_p(t) \equiv \tau_c(t) = \begin{cases} \tau_{\text{ns}}(t), & v_{\text{rel}} = 0 \text{ and } |\tau_{\text{ns}}(t)| \le \tau_{\max}(t), \\ \tau_{\max}(t)\,\text{sign}\big(v_{\text{rel}}(t)\big), & \text{otherwise}, \end{cases} \tag{12.6}$$

with $v_{\text{rel}} = r_{p,\text{eff}}\omega_p - r_{s,\text{eff}}\omega_s$ (Section 11.1).

Once τ_p is selected, τ_s follows immediately from Equation (12.5).

12.4 Rotational Dynamics

The rotational equations of motion are Equation (9.1) and Equation (9.2). Written as first-order ODEs:

$$\dot{\omega}_p = \frac{1}{I_p}\left(\tau_{\text{eng}}(t) - \tau_p(t)\right), \tag{12.7}$$

$$\dot{\omega}_s = \frac{1}{I_s}\left(\tau_s(t) - \tau_{\text{load}}(t)\right). \tag{12.8}$$

Here $\tau_p(t)$ is given by the traction-limited rule Equation (12.6), and $\tau_s(t)$ is then computed from Equation (12.5).

12.5 Axial Shift Dynamics

The axial shift acceleration is given by Equation (8.37):

$$\ddot{s} = \frac{\Big(F_{p,\text{ax}}(s,\omega_p) + F_{c,p}(s,\omega_p)\Big) - \Big(F_{s,\text{ax}}(s,\tau_s) + F_{c,s}(s,\omega_s)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}. \tag{12.9}$$

In this assembled model, the secondary torque input appearing in $F_{s,\text{ax}}(s,\tau_s)$ is evaluated by Equation (12.5), where τ_p is selected by the traction rule Equation (12.6).

12.6 Closed Model (What Is Computed Each Step)

Given the state $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s})$ at time t:

1. Compute geometry from s: $r_{p,\text{eff}}(s)$, $r_{s,\text{eff}}(s)$, $\phi_p(s)$, $\phi_s(s)$, etc.
2. Compute ratio and ratio rate: $R(s)$ from Equation (12.2), and $\dot{R}(s,\dot{s})$ from Equation (12.3).
3. Compute the no-slip torque demand τ_{ns} from Equation (12.4).
4. Compute traction capacity $\tau_{\max}$ from Equation (10.36) (primary and secondary) and apply the chosen limiting rule.
5. Select the transmitted coupling torque τ_p using Equation (12.6).
6. Map to the secondary torque τ_s using Equation (12.5).

7. Evaluate the ODE right-hand-sides: Equation (12.7), Equation (12.8), and Equation (12.9).

Therefore the complete nonlinear ODE system is

$$\boxed{\begin{aligned} \dot{\omega}_p &= \frac{1}{I_p}\left(\tau_{\text{eng}}(t) - \tau_p\right), \\ \dot{\omega}_s &= \frac{1}{I_s}\left(\tau_s - \tau_{\text{load}}(t)\right), \\ \dot{s} &= \dot{s}, \\ \ddot{s} &= \frac{\Big(F_{p,\text{ax}}(s,\omega_p) + F_{c,p}(s,\omega_p)\Big) - \Big(F_{s,\text{ax}}(s,\tau_s) + F_{c,s}(s,\omega_s)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}, \end{aligned}} \tag{12.10}$$

with the algebraic closures

$$R = R(s), \quad \dot{R} = \dot{R}(s,\dot{s}), \quad \tau_p \text{ from Equation (12.6)}, \quad \tau_s = R\,\tau_p.$$

This completes the closed CVT dynamic model.

A Belt Length Small-Angle Approximation

This section quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models.

A.1 Standard Approximation

Many treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which leads to $\lambda \approx 0$ and therefore

A.2 Error Analysis

B Common Ramp Profile Designs

This section catalogs several common ramp profile parameterizations used in CVT primary pulley actuation.

B.1 Linear Ramp

The simplest ramp profile is a linear relationship:

$$r_f(s) = r_{f,0} + k_r \cdot s,$$

where k_r is the ramp slope (constant) and $r_{f,0}$ is the initial flyweight radius.

The corresponding ramp angle is constant:

$$\alpha_p(s) = \tan^{-1}(k_r).$$

Characteristics:

- Uniform force multiplication across the shift range
- Simple to manufacture and analyze
- May not provide optimal shift feel or calibration flexibility

B.2 Quadratic Ramp

B.3 Piecewise-Linear Ramp

B.4 Custom Profiles from Lookup Tables

C Numerical Methods Summary

This section summarizes the numerical solution methods used throughout this work.

C.1 Root Finding for Belt-Length Constraint

The belt-length constraint Equation (7.12) is solved using Brent's method, a robust bracketed root-finding algorithm combining bisection, secant, and inverse quadratic interpolation.

C.2 ODE Integration for Dynamic Simulation

D Secondary Capacity Closure

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